

GUNUNG LUMUT LANDSCAPE PROJECT

Project title	Gunung Lumut Landscape Project
Project ID	5873
Crediting period	01-Januari-2023 to 31-Desember 2072
Project lifetime	01 January 2023 – 31 December 2072; 50 years lifetime
(CCB) GHG accounting period	01 January 2023 – 31 December 2072; 50 years total period
Original date of issue	<i>For pipeline listing, DD-Month-YYYY is the date of submission</i> <i>For registration, DD-Month-YYYY is the date the project description was completed following the completion of the audit</i>
Most recent date of issue	<i>DD-Month-YYYY is the date on which the document was most recently submitted</i>
Version	1
VCS Standard version	VCS Version 4.4
CCB Standards version	CCB Version 3.0
Project location	East Kalimantan Province, Indonesia
Project proponent(s)	PT Telaga Mas Kalimantan
Validation/verification body	<i>Organization and contact name with email address and phone number</i>
History of CCB status	<i>Issuance date(s) of earlier validation statements, dates of previous attempts at validation, etc.</i>
Gold Level criteria	The project supports in assisting communities and biodiversity in adapting to impact of climate change through measures needed and designed for adaptation such as vulnerability index measurement and design of

	adaptation activities in order to achieve economic, social, income, ecosystem, and land resilience. Which will be explained in chapter 3 (GL 1). The project will aim to preserve forest and customary identity of the community through engagements with the Mului community located within the project area. Identify the short and long-term community benefits through various beneficial activities that could be provided for the community. The project will establish cooperative institutions and community-based companies for improving the economic sustainability of the community, identify marginalized and/or vulnerable community groups, and provide equal opportunities for women on development of community-based business. The project will also aim to create democratic system/member participation in decision making process, raising awareness of activities to communicate the benefits, costs, and risks to the community, and also create community enterprise governance and structure. There will also be training for welfare improvement, education programs, gender-based development, etc to improve the capacity and competency of smallholders/community members. This is explained in chapter 4 (GL 2). Maintenance and enhancement of high biodiversity conservation priority in GLLP will be implemented through : identification and management of high biodiversity conservation priority. And monitoring of habitat, population, targeted group (exotic species, primate group, bird community, small mammals, etc.), and specific purpose (plant-animals off take), and monitoring of trigger species population for the impacts of the project on biodiversity. Which can refer to chapter 5 (GL 3).
Expected verification schedule	<i>If known, provide the expected schedule for initial verification</i>
Prepared by	PT Telaga Mas Kalimantan, PT Wanantara Alesha Sadhana

CONTENTS

Update the page numbers of the table of contents below upon completion of the project description.

1	SUMMARY OF PROJECT BENEFITS	4
1.1	Unique Project Benefits	4
1.2	Standardized Benefit Metrics	6
2	PROJECT DETAILS	10
2.1	Project Goals, Design and Long-Term Viability	10
2.2	Without-project Land Use Scenario and Additionality	52
2.3	Safeguards and Stakeholder Engagement	53
2.4	Management Capacity	55
2.5	Legal Status and Property Rights	55
2.6	Additional Information Relevant to the Project	69
3	CLIMATE	70
3.1	Application of Methodology	70
3.2	Quantification of Estimated GHG Emission Reductions and Removals	86
3.3	Monitoring	87
3.4	Optional Criterion: Climate Change Adaptation Benefits	87
4	COMMUNITY	87
4.1	Without-Project Community Scenario	87
4.2	Net Positive Community Impacts	88
4.3	Other Stakeholder Impacts	88
4.4	Community Impact Monitoring	89
4.5	Optional Criterion: Exceptional Community Benefits	89
5	BIODIVERSITY	90
5.1	Without-Project Biodiversity Scenario	90
5.2	Net Positive Biodiversity Impacts	90
5.3	Offsite Biodiversity Impacts	91
5.4	Biodiversity Impact Monitoring	91
5.5	Optional Criterion: Exceptional Biodiversity Benefits	91
	APPENDIX 1: STAKEHOLDER DESCRIPTION TABLE	93
	APPENDIX 2: PROJECT ACTIVITIES AND THEORY OF CHANGE TABLE	93
	APPENDIX 3: PROJECT RISKS TABLE	93
	APPENDIX 4: COMMERCIAL SENSITIVE INFORMATION	93

1 SUMMARY OF PROJECT BENEFITS

1.1 Unique Project Benefits

Outcome or impact estimated by the end of project lifetime	Section reference
1) Climate The project will have an impact on climate change mitigation efforts, namely to reduce emissions by 47,053,072 tCO₂e with an area of 116,001 Ha (based on spatial delineation). This is derived from activities to avoid emissions originating from legal harvesting, illegal logging, deforestation into mining, and increasing carbon sequestration through planting and maintaining stands as well as maintaining the ecological functions of forests. In addition to having an impact on mitigation, the project will have an impact on efforts to adapt to climate change through increasing community welfare and efforts to protect/enhance biodiversity that is threatened by climate change.	3
2) Community The project is expected to increase community welfare through the establishment of economic institutions (cooperatives) in 24 villages. Through cooperatives, the community will have the ability to organize and increase their capacity and capability. In their business, the cooperatives will involve illegal loggers, marginalized groups and women's groups. The involvement of illegal loggers is expected to reduce illegal activities due to the preservation of forest activities. This is because the community has realized that the forest is their main source of income through optimizing Non-Timber Forest Products (NTFPs) in the form of certified rattan, forest honey farming, deer breeding, and conservation tourism. Meanwhile, the involvement of women's groups will optimize the role of women through agroforestry and Micro, Small and Medium Enterprises (MSMEs), so as to improve welfare quality indicators. The project highly respects the existence of indigenous or customary peoples. The project will increase the strength and influence of "adat" on the interaction of social life so that the preservation and sacredness of customs will be stronger. Apart from that, through the project, it is hoped that there will be a change in the pattern of land management from conventional to sustainable. Farmers will have the opportunity to produce organic fertilizer independently so that they can increase the selling price of agricultural/plantation products that are nowadays more acceptable to the market.	4

The project will also involve the Regional Government of Paser Regency to increase the legitimacy of the Government towards the community through the establishment of a more sustainable community-based company (Perseroan Daerah). Based on the activities carried out during the project period, the GLLP will contribute to improving the welfare of 3,074 people.	
3) Biodiversity The project will protect the habitat of endangered wildlife such as <i>Manis javanica</i> (pangolin), <i>Rhinoplax vigil</i> (helmeted hornbill), <i>Rhabdotorrhinus corrugatus</i> (black-crested hornbill), <i>Berenicornis comatus</i> (crested hornbill), <i>Hylobates muelleri</i> (Kelampiau), <i>Macaca fascicularis</i> (long tail monkey), <i>Macaca nemestrina</i> (beruk), <i>Cynogale bennettii</i> (Water Weasel). Wildlife protection is carried out through education and field practice for the community so that there will be no human-fauna conflicts, regular patrols involving the community, and expanding habitat through landscape connectivity and habitat restoration. The project will also protect and enrich various endangered species of flora such as <i>Shorea sumatrana</i> (Slooten) Desch, <i>Shorea pachyphylla</i> Ridl. ex Symington, <i>Shorea pauciflora</i> King, <i>Cinnamomum parthenoxylon</i> (Jack) Meisn., <i>Shorea ovata</i> Dyer, <i>Dracontomelon costatum</i> Blume, <i>Dipterocarpus hasseltii</i> Blume, and <i>Hopea ferrugineag</i> Parijis. Local community will be involved in all project activities including periodic standing inventory, making observation plots, procuring generative seeds, trial development research, planting vegetative seedlings trials, and planting/enrichment to increase species abundance.	5

1.2 Standardized Benefit Metrics

Category	Metric	Estimated by the end of project lifetime	Section reference
GHG emission reductions or carbon dioxide removals	<i>Net estimated removals in the project area, measured against the without-project scenario</i>	N/A	
	<i>Net estimated reductions in the project area, measured against the without-project scenario</i>	47,053,072 tCO ₂ e	3
Forest¹ cover	<i>For REDD² projects: Estimated number of hectares of reduced forest loss in the project area measured against the without-project scenario</i>	Under development	
	<i>For ARR³ projects: Estimated number of hectares of forest cover increased in the project area measured against the without-project scenario</i>	Under development	
Improved land management	<i>Number of hectares of existing production forest land in which IFM⁴ practices are expected to occur as a result of project activities, measured against the without-project scenario</i>	Under development	
	<i>Number of hectares of non-forest land in which improved land management practices are expected to occur as a result of project activities, measured</i>	N/A	

¹ Land with woody vegetation that meets an internationally accepted definition (e.g., UNFCCC, FAO, or IPCC) of what constitutes a forest, which includes threshold parameters, such as minimum forest area, tree height and level of crown cover, and may include mature, secondary, degraded and wetland forests (VCS Program Definitions)

² Reduced emissions from deforestation and forest degradation (REDD) - Activities that reduce GHG emissions by slowing or stopping conversion of forests to non-forest land and/or reduce the degradation of forest land where forest biomass is lost (VCS Program Definitions)

³ Afforestation, reforestation and revegetation (ARR) - Activities that increase carbon stocks in woody biomass (and in some cases soils) by establishing, increasing and/or restoring vegetative cover through the planting, sowing and/or human-assisted natural regeneration of woody vegetation (VCS Program Definitions)

⁴ Improved forest management (IFM) - Activities that change forest management practices and increase carbon stock on forest lands managed for wood products such as saw timber, pulpwood, and fuelwood (VCS Program Definitions)

Category	Metric	Estimated by the end of project lifetime	Section reference
	<i>against the without-project scenario</i>		
Training	<i>Total number of community members who are expected to have improved skills and/or knowledge resulting from training provided as part of project activities</i>	Under development	
	<i>Number of female community members who are expected to have improved skills and/or knowledge resulting from training as part of project activities</i>	Under development	
Employment	<i>Total number of people expected to be employed in project activities⁵, expressed as number of full-time employees⁶</i>	Under development	
	<i>Number of women expected to be employed as a result of project activities, expressed as number of full-time employees</i>	Under development	
Livelihoods	<i>Total number of people expected to have improved livelihoods⁷ or income generated as a result of project activities</i>	Under development	
	<i>Number of women expected to have improved livelihoods or income generated as a result of project activities</i>	Under development	

⁵ Employed in project activities means people directly working on project activities in return for compensation (financial or otherwise), including employees, contracted workers, sub-contracted workers and community members that are paid to carry out project-related work.

⁶ Full time equivalency is calculated as the total number of hours worked (by full-time, part-time, temporary and/or seasonal staff) divided by the average number of hours worked in full-time jobs within the country, region or economic territory (adapted from the UN System of National Accounts (1993) paragraphs 17.14[15.102];[17.28])

⁷ Livelihoods are the capabilities, assets (including material and social resources) and activities required for a means of living (Krantz, Lasse, 2001. The Sustainable Livelihood Approach to Poverty Reduction. SIDA). Livelihood benefits may include benefits reported in the Employment metrics of this table.

Category	Metric	Estimated by the end of project lifetime	Section reference
Health	<i>Total number of people for whom health services are expected to improve as a result of project activities, measured against the without-project scenario</i>	Under development	
	<i>Number of women for whom health services are expected to improve as a result of project activities, measured against the without-project scenario</i>	Under development	
Education	<i>Total number of people for whom access to, or quality of, education is expected to improve as result of project activities, measured against the without-project scenario</i>	Under development	
	<i>Number of women and girls for whom access to, or quality of, education is expected to improve as result of project activities, measured against the without-project scenario</i>	Under development	
Water	<i>Total number of people who are expected to experience increased water quality and/or improved access to drinking water as a result of project activities, measured against the without-project scenario</i>	Under development	
	<i>Number of women who are expected to experience increased water quality and/or improved access to drinking water as a result of project activities, measured against the without-project scenario</i>	Under development	

Category	Metric	Estimated by the end of project lifetime	Section reference
Well-being	<i>Total number of community members whose well-being⁸ is expected to improve as a result of project activities</i>	Under development	
	<i>Number of women whose well-being is expected to improve as a result of project activities</i>	Under development	
Biodiversity conservation	<i>Expected change in the number of hectares managed significantly better by the project for biodiversity conservation⁹, measured against the without-project scenario</i>	Under development	
	<i>Expected number of globally Critically Endangered or Endangered species¹⁰ benefiting from reduced threats as a result of project activities¹¹, measured against the without-project scenario</i>	Under development	

⁸ Well-being is people's experience of the quality of their lives. Well-being benefits may include benefits reported in other metrics of this table (e.g. Training, Employment, Livelihoods, Health, Education and Water), and may also include other benefits such as strengthened legal rights to resources, increased food security, conservation of access to areas of cultural significance, etc.

⁹ Managed for biodiversity conservation in this context means areas where specific management measures are being implemented as a part of project activities with an objective of enhancing biodiversity conservation, e.g. enhancing the status of endangered species

¹⁰ Per IUCN's Red List of Threatened Species

¹¹ In the absence of direct population or occupancy measures, measurement of reduced threats may be used as evidence of benefit

2 PROJECT DETAILS

2.1 Project Goals, Design and Long-Term Viability

2.1.1 Summary Description of the Project (VCS, 3.2, 3.6, 3.10, 3.11, 3.13, 3.14; CCB, G1.2)

Gunung Lumut Landscape Project (“GLLP”), located in Paser Regency, East Kalimantan Province, Indonesia, is one of the world's remaining tropical forests (See Figure 1). Paser Regency is directly adjacent to North Penajam Paser Regency, where the Capital City of Nusantara (New State Capital of Indonesia/”IKN”) is located. The regency will be a buffer zone for the IKN environment¹² and since GLLP is located approximately $\pm$ 75.60 km from IKN, its location is important for IKN (Figure 2). IKN has been proclaimed to replace DKI Jakarta as the capital of the Republic of Indonesia with the concept of Forest City. It is designed to have high natural conditions (creating 75 % of green area) with low carbon emission¹³

GLLP aims to protect and restore $\pm$ 116,001 Ha of concession consisting of predominantly lowland forest with fragmented forest condition, and the main Watersheds (DAS), namely the Telake River in the North-Western part of the project area and the Kendilo River in the West, South, and Eastern part of the project area. GLLP consists of concession areas for production forests (Unit I) and protected forests (Unit II). The Unit I project area has obtained a Forest Management Business License for Production Forest (“PBPH-HP”) which for Business As Usual (BAU) is given permit to harvest timber. The Unit II project area has obtained principal approval for a Forest Utilization Business Permit for Protected Forest (“PBPH-HL”). The protected forest in the project area is the Gunung Lumut Protected Forest (“HL Gunung Lumut”) which is the sacred land¹⁴ and cultural center of the Dayak Paser indigenous or customary people. HL Gunung Lumut is located in the middle of the production forest concession areas (Unit I) which also functions as the utilization block of the Gunung Lumut-Gunung Ketam landscape (see Figure 1). Within the project zone of GLLP, there is a customary forest area where the Mului indigenous or customary people live, which has received recognition from the Government of the Republic of Indonesia (the Ministry of Environment and Forestry (“KLHK”) as the Mului Customary Forest.

¹² Adinugroho WC, Prasetyo LB, Kusmana C, Krisnawati H. Tracking environmental quality of Indonesia's new capital city and its surrounding area. In: IOP Conference Series: Earth and Environmental Science 2022 (Vol. 950, No. 1, p. 012077). IOP Publishing.

¹³ <https://ikn.go.id/>

¹⁴ Gunung Lumut is the center of religious rituals native to the Dayak Paser customary people, namely the Kaharingan religion. The Mului customary people believe that Mount Lumut is a ladder where before the inner spirit goes to the final world, it will temporarily stop at Mount Lumut. Followers of the Kaharingan religion believe that when someone dies, the *sensinu* (shadow spirit) must be brought to Mount Lumut, and a procession is carried out for seven days. And on the 7th day, the *juus* spirit (inner spirit) will leave and finally live forever in heaven.

Before the project started, the areas for this project were designated to be production forests and protected forests. For the Gunung Lumut Protected Forest area (Unit II), the provincial government carried out the site management, followed by a policy change where the management is then taken over by a site-level forest management unit known as Kesatuan Pengelolaan Hutan (“KPH”). Currently, the HL Gunung Lumut is managed under the Telake forest management unit (KPH Telake) authority, where Unit II is included in the utilization Zone.

Aligning with the multi-business forestry concept regulated by the Ministry of Environment and Forestry (Regulation No. 8 of 2021), the authority to manage the protected forests is now allowed to be managed by private parties. GLLP has initiated a change in business activities in the production forest area (Unit I) by including the ecosystem services for carbon sequestration and storage while also placing the HL Gunung Lumut area (Unit II) in a strategic position as a buffer zone for Unit I to form a compact landscape to prevent fragmentation of wildlife habitat. With the project’s intervention, the production forest area will be converted into a protected area while management of the protected forest area will be strengthened to ensure undisturbed ecological functions of the project area. Project interventions to increase productivity in protected forest areas are also carried out through ecosystem services produced by forests (e.g., carbon) and is conducted with community participation to ensure achievement in sustainability. In a global challenge such as climate change, biodiversity loss, and socioeconomic disparities, "strong sustainability" offers a clear concept for a sustainable future. Natural capital, including biodiversity, ecosystems, and non-renewable resources, has an inherent value that cannot be entirely replaced or duplicated by man-made capital that demonstrates the interconnectedness and interdependence of natural and social systems, emphasizing the need to preserve ecological integrity¹⁵. Achieving "strong sustainability" requires a collective effort, transformative change, and a solid commitment to preserving ecosystems and natural resources for future generations.

GLLP Unit II is considered one of the centers for plant and wildlife biodiversity within Kalimantan Island (Tropenbos, 2009). Based on the survey results conducted by PT TMK in 2022, at least 1 (one) floral species were categorized as critically endangered, 8 (eight) as endangered, and 15 (fifteen) as vulnerable. The ecosystem of Gunung Lumut-Gunung Ketam and unit I areas were considered priority areas for species of Rangkong/Hornbills, which is categorized as critically endangered species in Indonesia. In terms of fauna species, there were 118 species of birds, 20 mammals, and 7 (seven) herpetofauna found in the project area. Among them, at least 2 (two) species were categorized as critically endangered, 7 (seven) as endangered, and 10 (ten) as vulnerable. Tropenbos, (2009) stated that the protected forest of Gunung Lumut is considered a forest ecosystem with a richer biodiversity than any other forests within Kalimantan Island.

In addition to its rich biodiversity, East Kalimantan is the largest coal producer in Indonesia according to the Ministry of Energy and Mineral Resources, 2020. The quality and calorie value of coal from East Kalimantan is considered above average, causing more coal from this province to

¹⁵ Giddings B, Hopwood B, O'brien G. Environment, economy and society: fitting them together into sustainable development. Sustainable development. 2002 Nov;10(4):187-96.

be exported than used to meet domestic energy needs. East Kalimantan contributed 268,449 tons or 47.9% of the total coal extracted in Indonesia and is Indonesia's largest coal exporting region. In 2020, this province exported 212.8 tons of coal abroad¹⁶. Mining contributes up to 70% of East Kalimantan's economy. Based on data from the Ministry of Energy and Mineral Resources (ESDM) as of November 2021, the number of Mining Business Permit (IUP) in East Kalimantan has reached 476 permits, which are divided into 437 IUP, 1 IUPK (Special Mining Business Permit), 11 KK (Contract of Work), and 37 PKP2B (Coal Mining Concession Work Agreement)¹⁷.

The potential for coal deposits under the forest stands in East Kalimantan poses a distinct threat of converting forests into mining. Deforestation caused by mining is a significant problem in East Kalimantan, Indonesia¹⁸. Coal mining, in particular, has been identified as a substantial cause of mining-related deforestation in the country¹⁹. Mining-related deforestation in East Kalimantan has significantly impacted local communities and the environment, including damage to livelihoods and forest ecosystems²⁰. This threat also occurs in the GLLP project area, especially Unit I, a production forest; where legally, the prevailing laws allow coal mining permits to be submitted on the area. Article 91 of Government Regulation of the Republic of Indonesia No. 23 of 2021 states that using forest areas for development purposes other than forestry activities is possible, including mining. The maximum quota that can be considered is 10% (ten percent) of the effective area of each Forest Management Business License ("PBPH") so that there is an area of $\pm 6,861.8$ Ha in GLLP Unit I that can be converted into mining (Figure 3). Based on this allocation, currently, there are areas for which a decree on the approval on the use of forest areas has been issued for mining production operations and its supporting facilities.

The GLLP project area also has other challenges and threats. Illegal activities such as illegal logging have occurred within the last few decades. Furthermore, there are illegal conversions of forests into palm and rubber plantations by the local communities. In fact, between 2000-2005, illegal logging in GLLP Unit II was intensive even though a ministerial decree (Decree of the Ministry of Forestry No. 24/Kpts/Um/1983) have passed declaring the Gunung Lumut forest to be a protected forest area prohibiting all logging activities (TBI, 2004).

Lack of awareness and lack of opportunities to improve the well-being of community members have caused illegal activities such as illegal logging and mining to happen. The main challenge for managing protected forest areas is to achieve consensus in formulating strategies that follow the area's environment, economy, and socio-culture principles. For this, the management of natural

¹⁶ <https://www.forestdigest.com/detail/2087/batu-bara-kalimantan-timur#:~:text=Kalimantan%20Timur%20juga%20merupakan%20daerah,batu%20bara%20ke%20luar%20negeri.>

¹⁷ <https://infotambang.id/perizinan/peta-perizinan-tambang-kalimantan-timur/>

¹⁸ Giljum S, Maus V, Kuschig N, Luckeneder S, Tost M, Sonter LJ, Bebbington AJ. A pantropical assessment of deforestation caused by industrial mining. *Proceedings of the National Academy of Sciences*. 2022 Sep 20;119(38):e2118273119.

¹⁹ <https://phys.org/news/2022-09-tropical-deforestation-attributable-industrial-indonesia.html>

²⁰ <https://news.mongabay.com/2022/06/in-indonesian-borneo-a-succession-of-extractive-industries-multiplies-impacts-social-fractures/>

resources could not be narrowly defined as only ecosystem-based biodiversity conservations but must also be interpreted as sustainable management of natural resources.

The project activities are designed to maximize the capability of project proponent to restore ecological functions within the project area, increase ecosystem productivity, and building relationships with stakeholders within the landscape for the sustainability of the project. The activities to be carried out in the project area include monitoring and area patrols, conversion of logged to protected forest, preventing conversion to mining areas, increasing forest cover through planting and maintenance activities, as well as community development. Community development and participation are needed so the positive impact of this project can continue in the future even when project proponent is no longer present in the area (project permanency). It is estimated that the GLLP will generate about $\pm 941,061$ tCO₂e annually or total of $\pm 47,053,072$ tCO₂e for 50 years from climate change mitigation benefits.

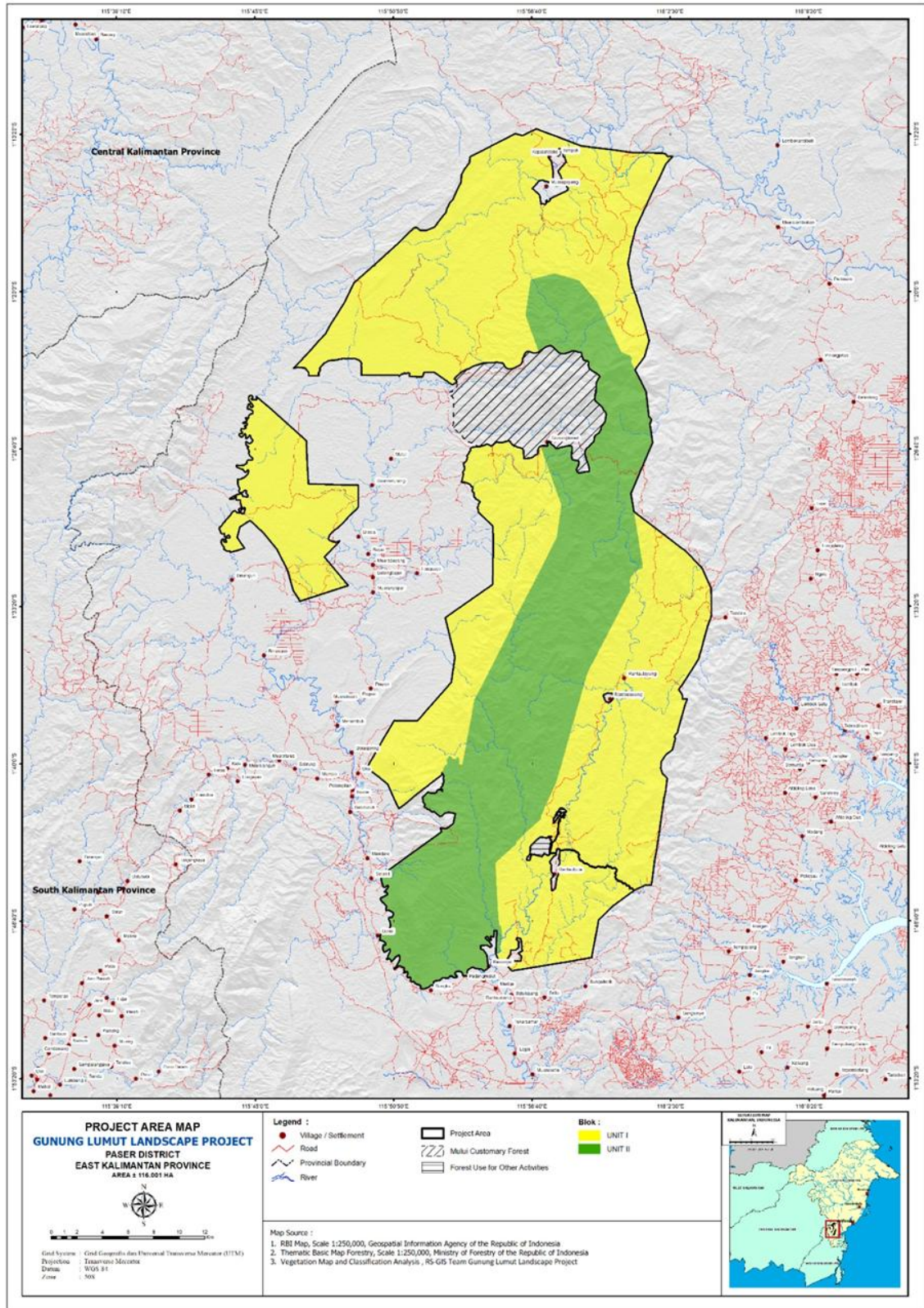


Figure 1. Map of GLLP Project Area

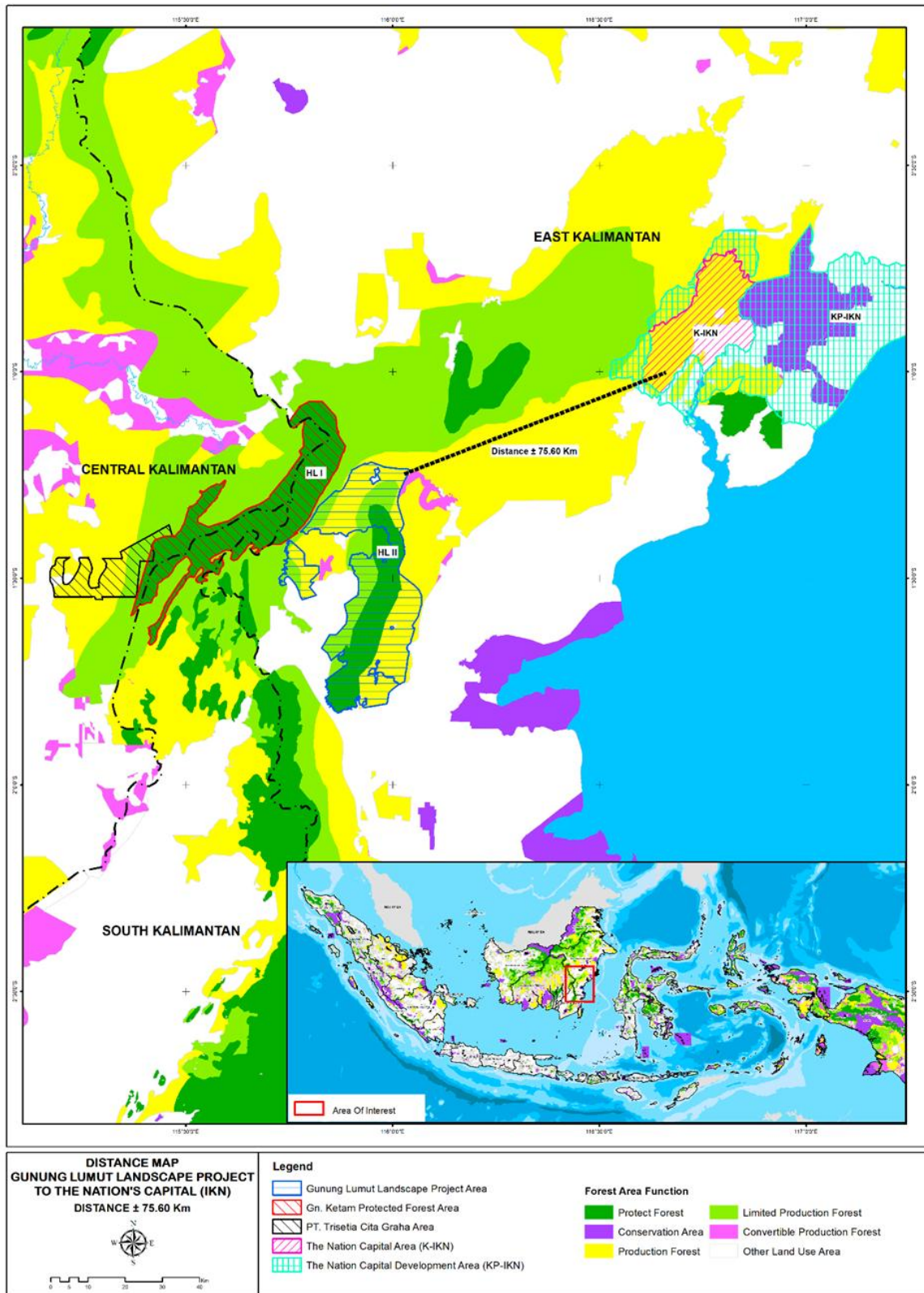


Figure 2. Distance Between GLLP and IKN

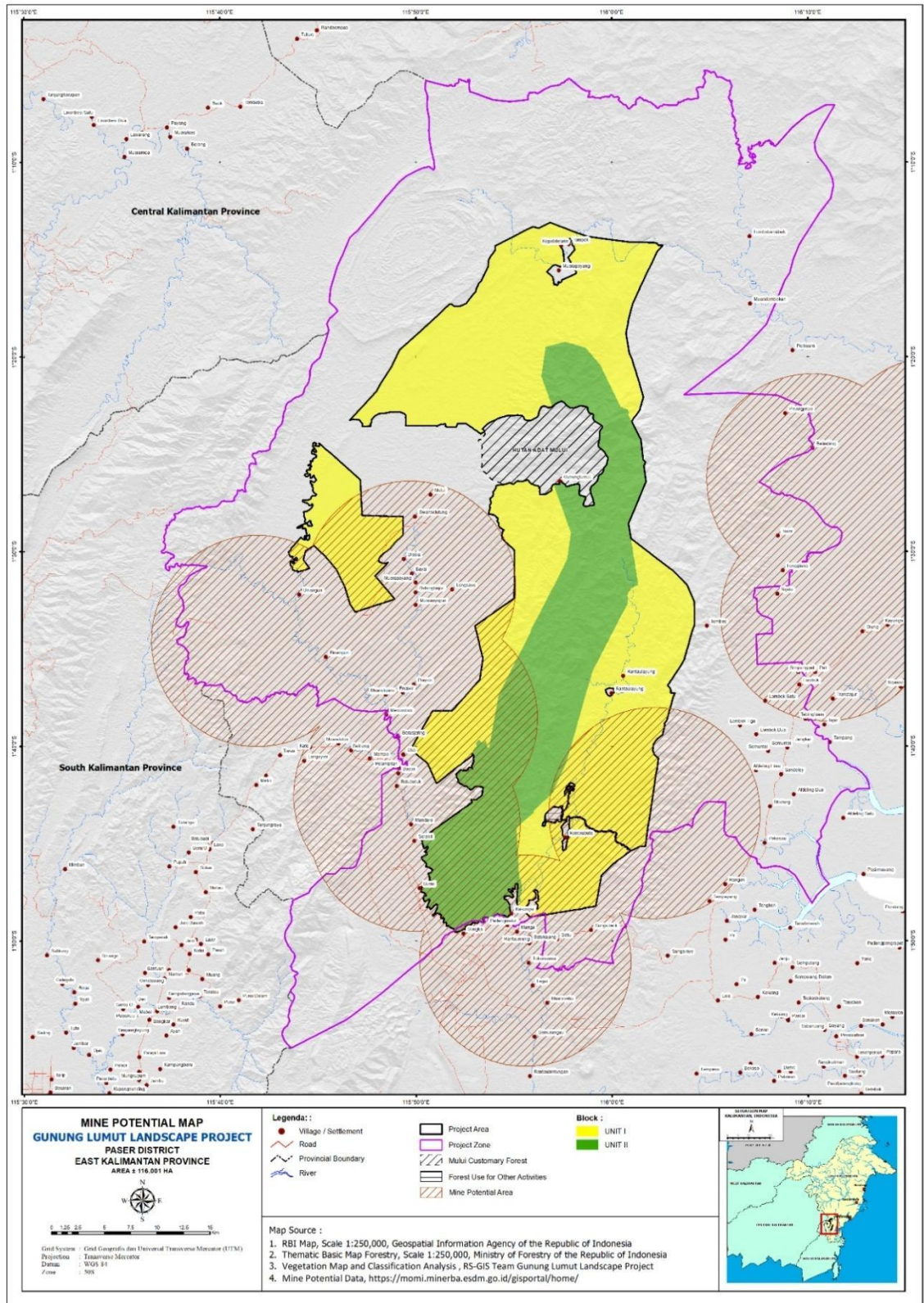


Figure 3. Map of Potential Mining Area in GLLP

2.1.2 Audit History (VCS, 4.1)

This document is the first version of the GLLP Project. Therefore currently have no audit history.

2.1.3 Sectoral Scope and Project Type (VCS, 3.2)

Sectoral scope	14: Agriculture, forestry, and other land use
AFOLU project category ²¹	IFM (Improved Forest Management), ARR (Afforestation, Reforestation, and Revegetation), and REDD (Reducing Emissions from Deforestation and Degradation).
Project activity type	The project will combine the following activities: Improved Forest Management (IFM) in unit I: conversion from logged to protected forest; Afforestation, Reforestation, and Revegetation (ARR) in unit I and II; and Reduced Emissions from Deforestation and Degradation (REDD) : Avoid Planned Deforestation in Unit I and Avoid Unplanned Deforestation in Unit II.

2.1.4 Project Eligibility (VCS, 3.1, 3.6, 3.8, 3.18, 4.1; CCB Program Rules, 4.2.4, 4.6.4)

This project is eligible for registration under both the Verified Carbon Standard (VCS) and the Climate, Community and Biodiversity (CCB) Standards as a multi-category AFOLU initiative. The GLLP project shall integrate activities across IFM (Improved Forest Management), ARR (Afforestation, Reforestation, and Revegetation), and REDD (Reducing Emissions from Deforestation and Degradation). All of which area encompassed within the VCS Scope 14: Agriculture, Forestry, and Other Land Use.

2.1.5 Transfer Project Eligibility (VCS, 3.23, Appendix 2)

The project activities do not fall under the criteria of a transfer project. Furthermore, the project will not result in double counting, in line with the statement on double counting provided in Section 2.5 of this document.

2.1.6 Project Design (VCS, 3.6)

Indicate if the project has been designed as:

- ☒ Single location or installation
- ☐ Multiple locations or project activity instances (but not a grouped project)
- ☐ Grouped project

2.1.6.1 Eligibility Criteria for Grouped Projects (VCS, 3.6; CCB, G1.14)

²¹ See Appendix 1 of the VCS Standard

Not applicable. The project is not a grouped project

2.1.7 Project Proponent (VCS, 3.7; CCB, G1.1)

Organization name	PT Telaga Mas Kalimantan ("PT TMK")
Contact person	Abdul Karim Sutanto
Title	Director
Address	Central Office: Graha Mitra, 9th Floor, Jl. Jend. Gatot Subroto Kav.21, Jakarta Selatan, DKI Jakarta 12930, Indonesia Site: Jl. Padat Karya No. 9 RT 16 Batu Sopang, Paser, Kalimantan Timur 76752
Telephone	+622139733397
Email	Abdul.Sutanto@indikanature.co.id

2.1.8 Other Entities Involved in the Project

Organization name	PT Indika Multi Properti ("PT IMP")
Role in the project	As the holding company of PT TMK as well as the guarantor for the funding of GLLP project
Contact person	Abdul Karim Sutanto
Title	Director
Address	Graha Mitra, 9th Floor, Jl. Jend. Gatot Subroto Kav.21, Jakarta Selatan, DKI Jakarta 12930, Indonesia
Telephone	+622139733397
Email	IMP@indikanature.co.id

Organization name	PT Wanantara Alesha Sadhana ("PT WAS")
Role in the project	As expert in the field of carbon accounting and biodiversity of the GLLP Project
Contact person	Dwi Shanty Apriliani Gunadi
Title	Head of Carbon Certification
Address	Jl. Yasmin Raya No.11-A, RT.10/RW.08, Curugmekar, Kec. Bogor Bar., Kota Bogor, Jawa Barat 16113

Telephone	+622518345920
Email	shanty@wanantara.com

2.1.9 Project Ownership (VCS, 3.2, 3.7, 3.10; CCB, G5.8)

Project ownership rests with PT TMK, a legally registered company under Indonesian law. The GLLP project area consists of 2 unit areas of Unit I and Unit II. For the unit I project area, PT TMK holds exclusive rights to manage approximately 82,805 Ha within KPHP Telake and KPHP Bongan, Paser Regency, East Kalimantan Province, as granted through the Forest Timber Product Utilization Business Permit in Natural Forest (IUPHHK-HA). These rights were extended by Ministerial Decree No. SK.34/Menlhk/Setjen/HPL.0/1/2020 dated 14 January 2020 and confirmed by Decree No. SK.782/MENLHK/SETJEN/HPL.0/9/2021 dated 16 September 2021.

As for Unit II, the project have obtained approval of licensing commitment to undertake forest utilization in protected forest Number: 10082311116401003 and have fulfill the requirements of the licensing commitment which is approval of the making of coordinates for the boundaries of prospective work area permit Number BA.47/BPHL.XI/P3HPL/KG/9/2023 to obtain the approval license for Forest Utilization in protected forest for Gunung Lumut Protected Forest area.

PT TMK is the sole project proponent with the authority to register and implement GLLP under the Verified Carbon Standard (VCS) and Climate, Community & Biodiversity Standards (CCB).

2.1.10 Project Start Date (VCS, 3.8)

Project start date	01-01-2023
Justification	As of the start date, mitigation measures have been implemented from meeting with mining company PT X and issued a notice letter to the Ministry of Energy and Mineral Resources in avoiding no more opening of land area for mining. As well as work plan (RKUPH) for for forestry multi-business plan specifically for planting and Intensive Silviculture have been issued.

2.1.11 Benefits Assessment and Project Crediting Period (VCS, 3.9; CCB, G1.9)

Crediting period	50 years
Start date of first or fixed	01-January-2023 to 01-January- 2072

crediting period	
CCB benefits assessment period	The CCB benefits will be sustained throughout the entire lifetime of the project, which in this case is 50 years.

2.1.12 Differences in Assessment/Project Crediting Periods (CCB, G1.9)

There are no difference between the GHG emissions accounting, the community, and/or biodiversity assessment and periods.

2.1.13 Project Scale and Estimated Reductions or Removals (VCS, 3.10)

☐ < 300,000 tCO₂e/year (project)

☒ ≥ 300,000 tCO₂e/year (large project)

Calendar year of crediting period	Estimated reductions or removals (tCO ₂ e)
01/01/2023 to 31/12/2023	296,372
01/01/2024 to 31/12/2024	350,677
01/01/2025 to 31/12/2025	430,534
01/01/2026 to 31/12/2026	480,294
01/01/2027 to 31/12/2027	534,835
01/01/2028 to 31/12/2028	1,590,033
01/01/2029 to 31/12/2029	1,575,146
01/01/2030 to 31/12/2030	1,665,591
01/01/2031 to 31/12/2031	1,708,293
01/01/2032 to 31/12/2032	1,681,645
01/01/2033 to 31/12/2033	844,394
01/01/2034 to 31/12/2034	857,069
01/01/2035 to 31/12/2035	914,204
01/01/2036 to 31/12/2036	867,081
01/01/2037 to 31/12/2037	906,346

Calendar year of crediting period	Estimated reductions or removals (tCO ₂ e)
01/01/2038 to 31/12/2038	962,000
01/01/2039 to 31/12/2039	948,576
01/01/2040 to 31/12/2040	917,590
01/01/2041 to 31/12/2041	1,088,938
01/01/2042 to 31/12/2042	1,102,235
01/01/2043 to 31/12/2043	1,178,133
01/01/2044 to 31/12/2044	1,182,895
01/01/2045 to 31/12/2045	1,055,845
01/01/2046 to 31/12/2046	1,115,534
01/01/2047 to 31/12/2047	974,444
01/01/2048 to 31/12/2048	941,356
01/01/2049 to 31/12/2049	969,281
01/01/2050 to 31/12/2050	951,266
01/01/2051 to 31/12/2051	775,797
01/01/2052 to 31/12/2052	836,747
01/01/2053 to 31/12/2053	768,268
01/01/2054 to 31/12/2054	733,303
01/01/2055 to 31/12/2055	778,967
01/01/2056 to 31/12/2056	771,451
01/01/2057 to 31/12/2057	817,601
01/01/2058 to 31/12/2058	851,204
01/01/2059 to 31/12/2059	842,346
01/01/2060 to 31/12/2060	852,192
01/01/2061 to 31/12/2061	870,595

Calendar year of crediting period	Estimated reductions or removals (tCO ₂ e)
01/01/2062 to 31/12/2062	876,439
01/01/2063 to 31/12/2063	887,996
01/01/2064 to 31/12/2064	884,186
01/01/2065 to 31/12/2065	925,055
01/01/2066 to 31/12/2066	877,245
01/01/2067 to 31/12/2067	910,301
01/01/2068 to 31/12/2068	928,516
01/01/2069 to 31/12/2069	920,049
01/01/2070 to 31/12/2070	873,841
01/01/2071 to 31/12/2071	996,503
01/01/2072 to 31/12/2072	983,865
Total Estimated ERRS	47,053,072
Total Number of Years	50
Average Annual ERRs	941,061

2.1.14 Physical Parameters (CCB, G1.3)

The project area is administratively located in Paser Regency, East Kalimantan province. The geographic coordinates are located at 116°04'13,39" - 115°43'26,55" E and 01°13'4,78" - 01°49'32,65" S. With a total area of 116,001 Ha that includes five districts, namely: Muara Komam, Long Kali, Long Ikis, Kuaro, and Batu Sopang .

1. Landcover

Based on the Planet Scope satellite image interpretation and landcover in 2022, the project area mainly consists of secondary dry land forest (95.331%), with a small portion of the area consisting of shrubs (3.422%), dry agricultural land mixed with shrubs (0.767%), water bodies (0.261%), open land areas (0.216%), and forest plantations (0.004%). Details of the distributions can be seen in the table and figure below (Table 1 and Figure 4).

Table 1. Landcover of the Project Area

No.	Landcover	Total	
		Ha	%
1	Secondary Dry Land Forest	110,584	95.331
2	Shrubs	3,970	3.422
3	Dry agricultural land mixed with shrubs	890	0.767
4	Bodies of water	302	0.261
5	Open Land Areas	250	0.216
6	Forest Plantations	4	0.004
Total		116,001	100.00

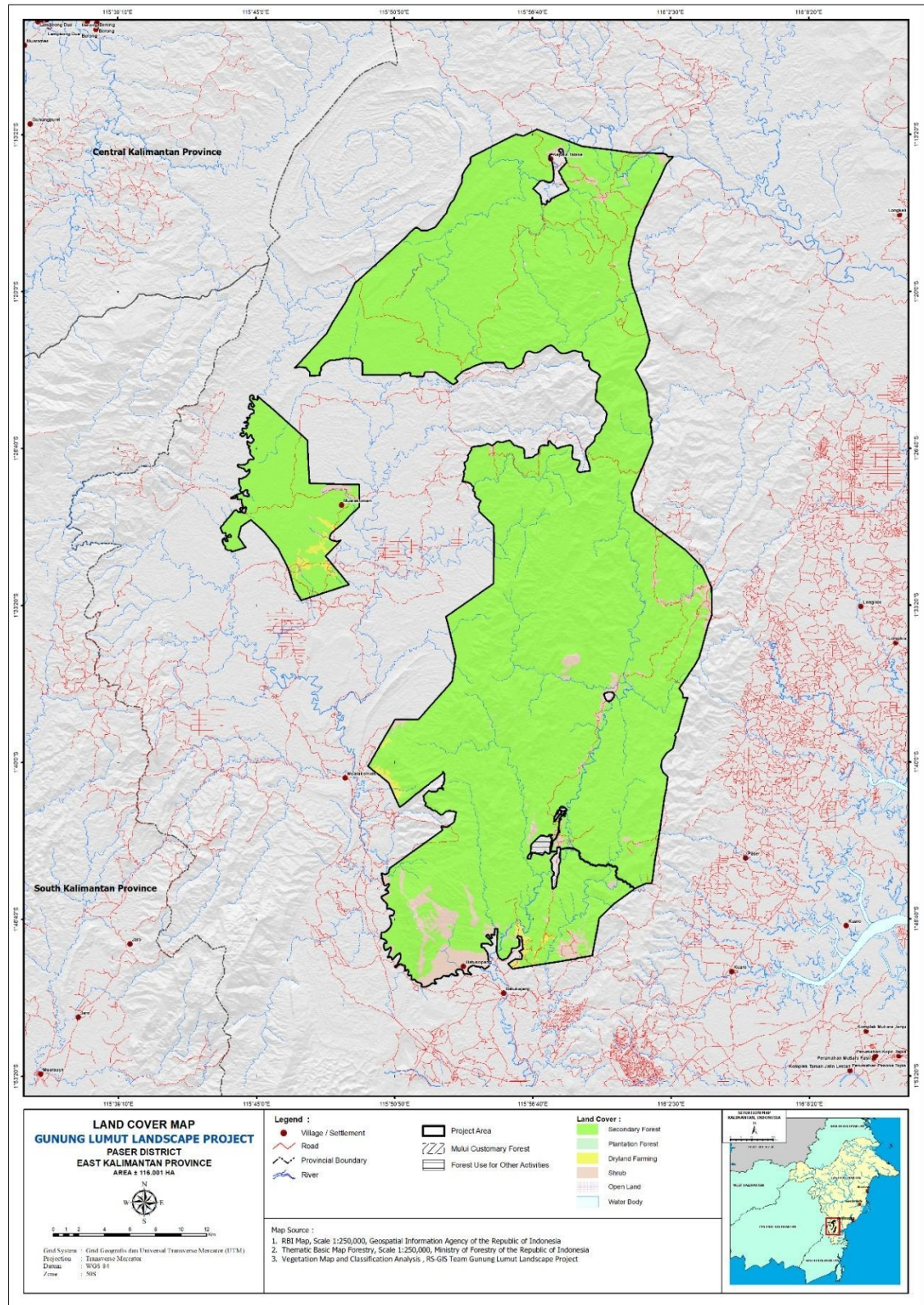


Figure 4. Landcover Map of Project Area

2. Topography

Slope Classification

The slope gradient within the project area are classified into 5 (five) classes, that is flat (0-8%), moderate (8-15%), rather steep (15-25%), steep (25-45%), and extremely steep (>45%). Based on spatial analysis, 33,710 Ha or 29.06% of the total project area are considered within the steep level. However, there are several locations that are flats such as in Batu Sopang, a highland area that potentially can be developed into seasonal agriculture land that can be managed without the need for significant investments. Other areas such as Muara Komam and Batu Sopang are regions that dominantly falls within the steep class category, hence, limiting factor for seasonal agriculture-based activities. Details of the slope gradient class and distribution can be seen in Table 2 and Figure 5 below.

Table 2. Slope Classification of the Project Area

No	Class	Area (Ha)	Percentage (%)
1	0 – 8%	16,614	14.32
2	8 – 15%	23,132	19.94
3	15 – 25%	27,704	23.88
4	25 – 45%	33,710	29.06
5	> 45%	14,840	12.80
Total		116,001	100.00

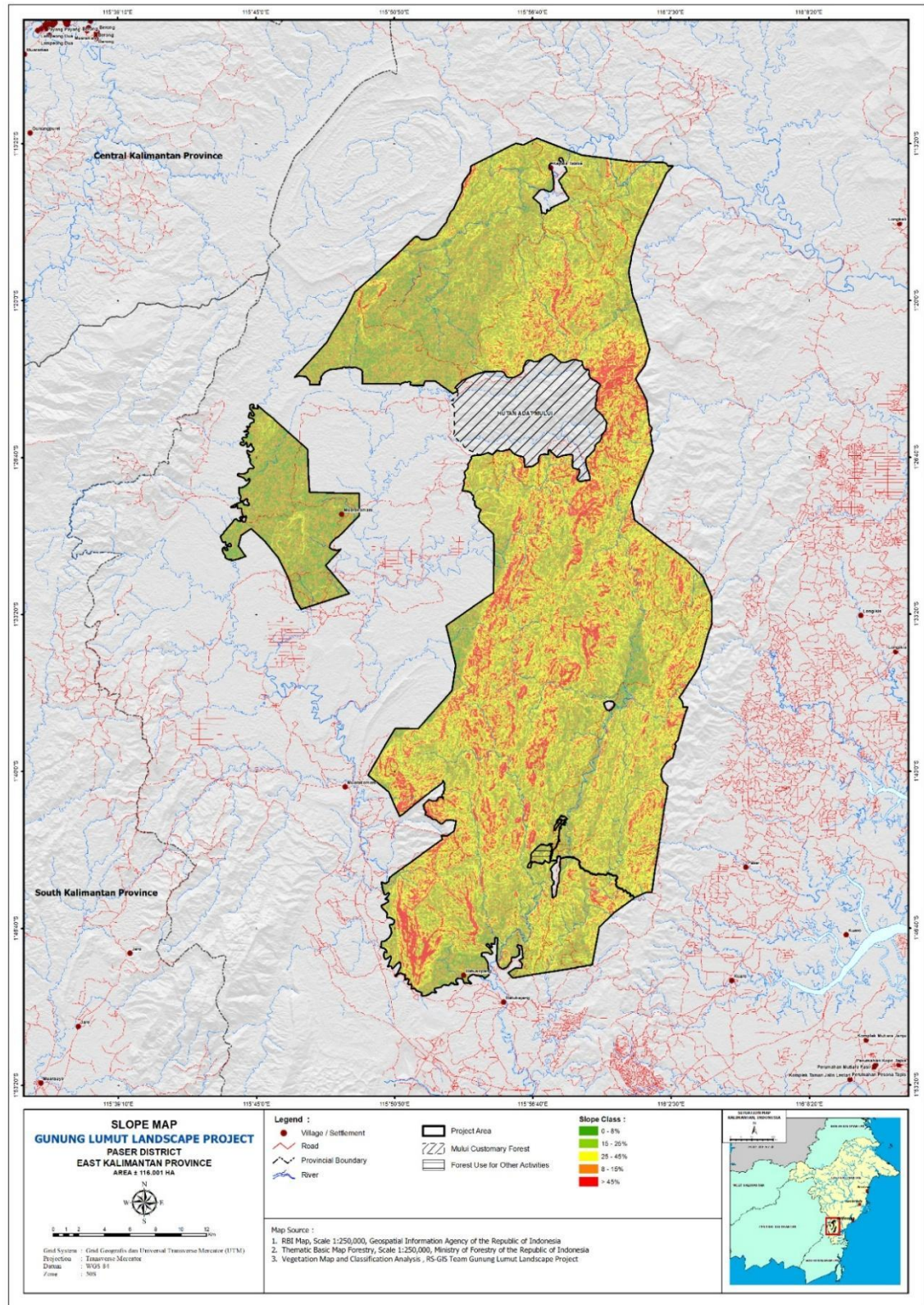


Figure 5. Map of Slope Gradient Distribution in Project Area

Elevation

The elevation within the project area can be classified into 6 (six) categories that are within the range classes of 0-200 masl, 200-400 masl, 400-600 asl, 600-800 masl, 800-1,000 masl, and 1,000-1,200 masl. The classes are categorized based on the needs of data for further analysis, mainly in classifying the ecosystem types within the area. Based on the analysis, the largest elevation class area within the project area is within the 200-400 masl elevation range with a total area of 47,292 Ha or 40.77% of the whole project area. The area with the highest elevation range in the project area is 1,000-1,200 masl with a total area of 223 Ha. Details and distribution can be seen in Table 3 and Figure 6.

Table 3. Elevation Class of The Project Area

No	Elevation Class	Area (Ha)	Percentage (%)
1	0 – 200	44,291	38.18
2	200 – 400	47,292	40.77
3	400 – 600	20,011	17.25
4	600 – 800	3,622	3.12
5	800 – 1.000	562	0.48
6	1.000 - 1.200	223	0.19
Total		116,001	100.00

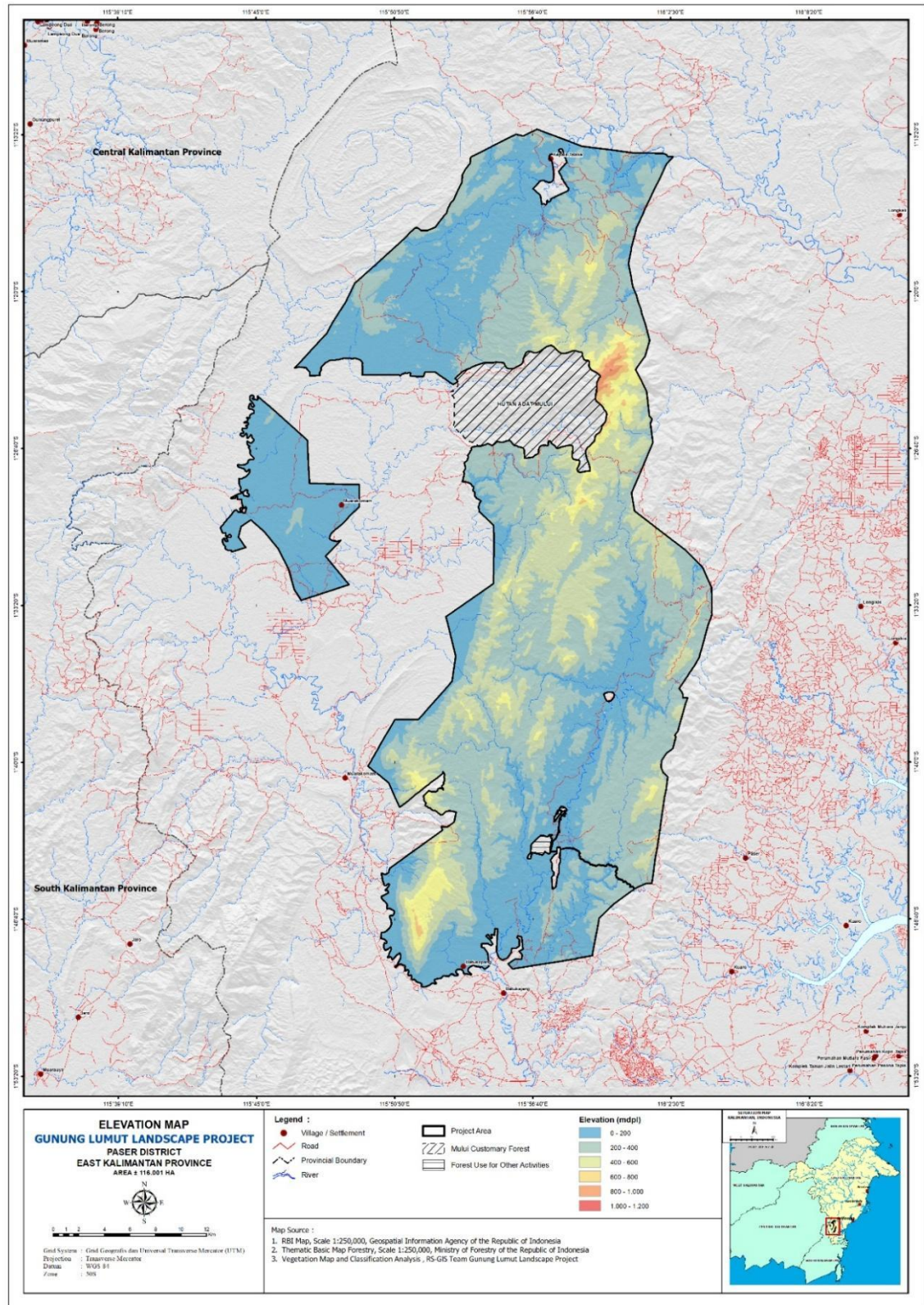


Figure 6. Distribution of Elevation Class

3. Soil Types and Land System

Land System

The land system is a combined information based on ecological principles connected with the types of rocks, hydroclimate, landform, soil types, and organisms, where the same environmental factor combinations will generate the same land system. Based on the analyzed results, the project area has 16 land systems where the largest area occupies the Maput Land System category²² located on rocky hills with moderate to slow soil permeability²³.

Soil Types

Soil as one of the biophysical elements that make up an area consists of several components, namely solid material (minerals), organic matter, water, and air. The soil types within the project area are categorized into two types, alluvial and podzolic soil types. The land area with alluvial soil type is 51,269 Ha and with podzolic soil type is 64,732 Ha. Alluvial soil type is usually found in lands that experience recent or frequent flooding, this soil type is considered young and has not experienced horizon differentiation²⁴. Most of the properties of alluvial soil type are from transported or deposited materials, therefore the properties vary depending on the parent materials deposited. Also, the distribution of alluvial soils is not affected by altitude or climate²⁵. Because alluvial soil properties are directly influenced by the original parent material, its fertility is also determined by the nature of the original material (Arabia et al. 2015). As regards podzolic soils, the soil type is formed due to heavy rainfall and very low temperatures. This soil type is also considered old mineral soil generally with a yellowish and reddish color. The color of the podzolic soil indicates a relatively low soil fertility (Agricultural Research and Development) (Refer to Table 4 and Figure 7).

Table 4. Soil Types in The Project Area

No	Soil Type	Area (Ha)	Percentage (%)
1	Alluvial	51,269	44.20
2	Podzolic	64,732	55.80
Total		116,001	100.00

²²The maput land system consists of irregular non-sedimentary rock hills with a slope of 0 to above 40%, dominant mineral rocks, namely sandstone, silt, mud and marl, rainfall 1600 – 4400 mm/year and the great soil group Hapludults and Dsytropepts (Muli Edwin, 2018)

²³ Edwin, M. Analisis Deskriptif Kondisi Vegetasi dan Sifat-sifat Tanah Pada Sistem Lahan Maput (MPT) dan Pendreh (PDH) di Hutan Wehea Kabupaten Kutai Timur.

²⁴ Hardjowigeno, S. 2003. Ilmu Tanah. Akademika Pressindo. Jakarta.

²⁵ Fiantis, Dian, et al. "Geochemical fingerprinting of volcanic soils used for wetland rice in West Sumatra, Indonesia." *Geoderma Regional* 10 (2017): 48-63.

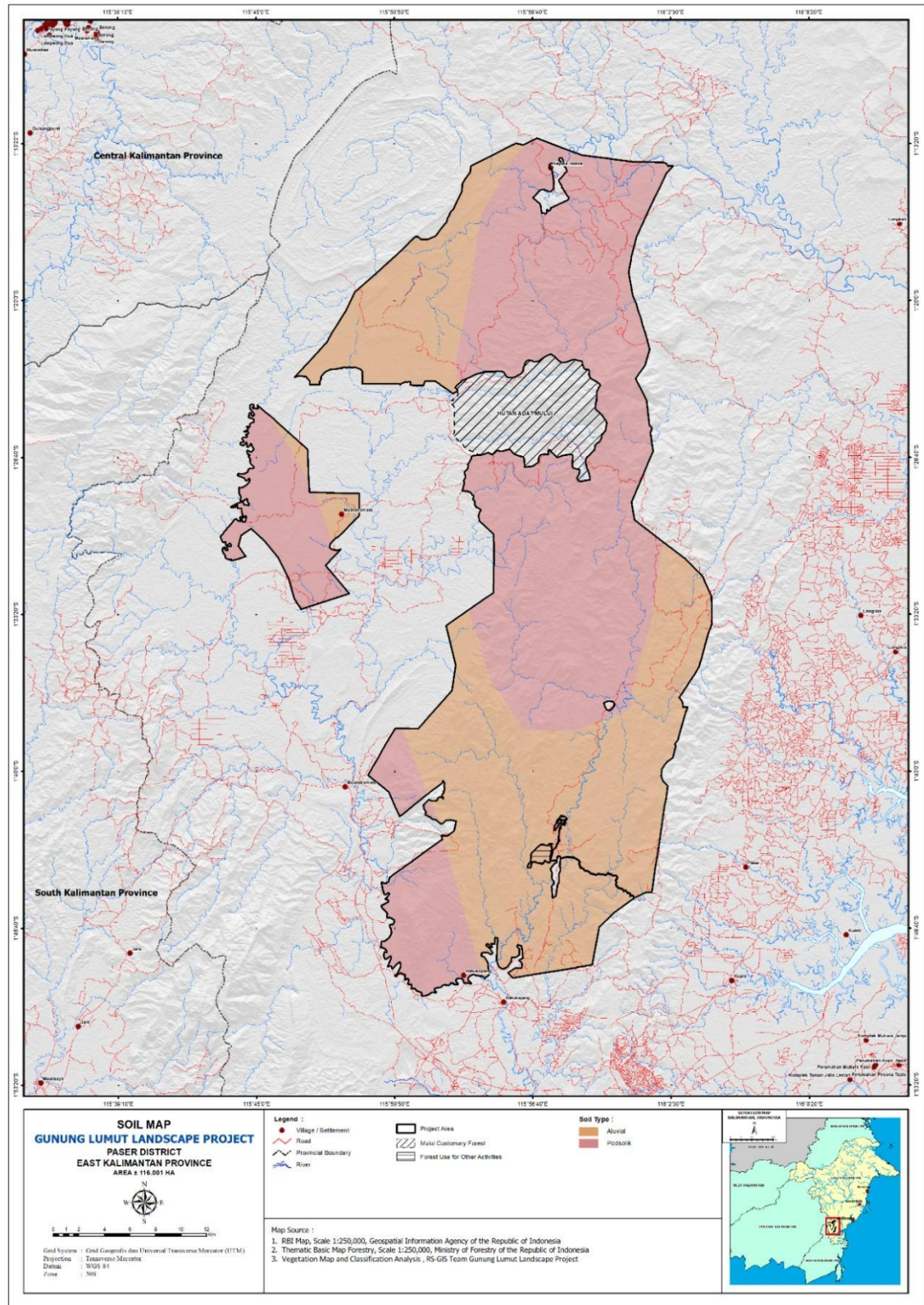


Figure 7. Map of Soil Type Distributions in Project Area

4. Climate

The project area is located in Paser Regency, East Kalimantan Province with a tropical climate that is highly influenced by changes in the wind passing through the site. In addition, other climate factors such as temperature and rainfall affect the changing weather conditions from year to year. The rainfall in Paser Regency in September 2022 was in the moderate category (50-75 mm) (Refer to Table 5 and Figure 8). Based on data source from the Department of Food Crops and Agriculture in Paser Regency in 2021, Long Kali District had the highest average rainfall of the other districts of the district with 279.61 mm and Kuaro District had the most frequent count of rainfall days per month with 215 days. Based on data from the Meteorological, Climatological, and Geophysical Agency in September 2022, the rainfall condition of the project area is categorized as low to moderate.

Table 5. Rainfall Conditions in Paser Regency

No	Rainfall (mm)	Area (Ha)	Percentage (%)
1	20 – 50	28,536	24.60
2	50 – 75	87,465	75.40
Total		116,001	100.00

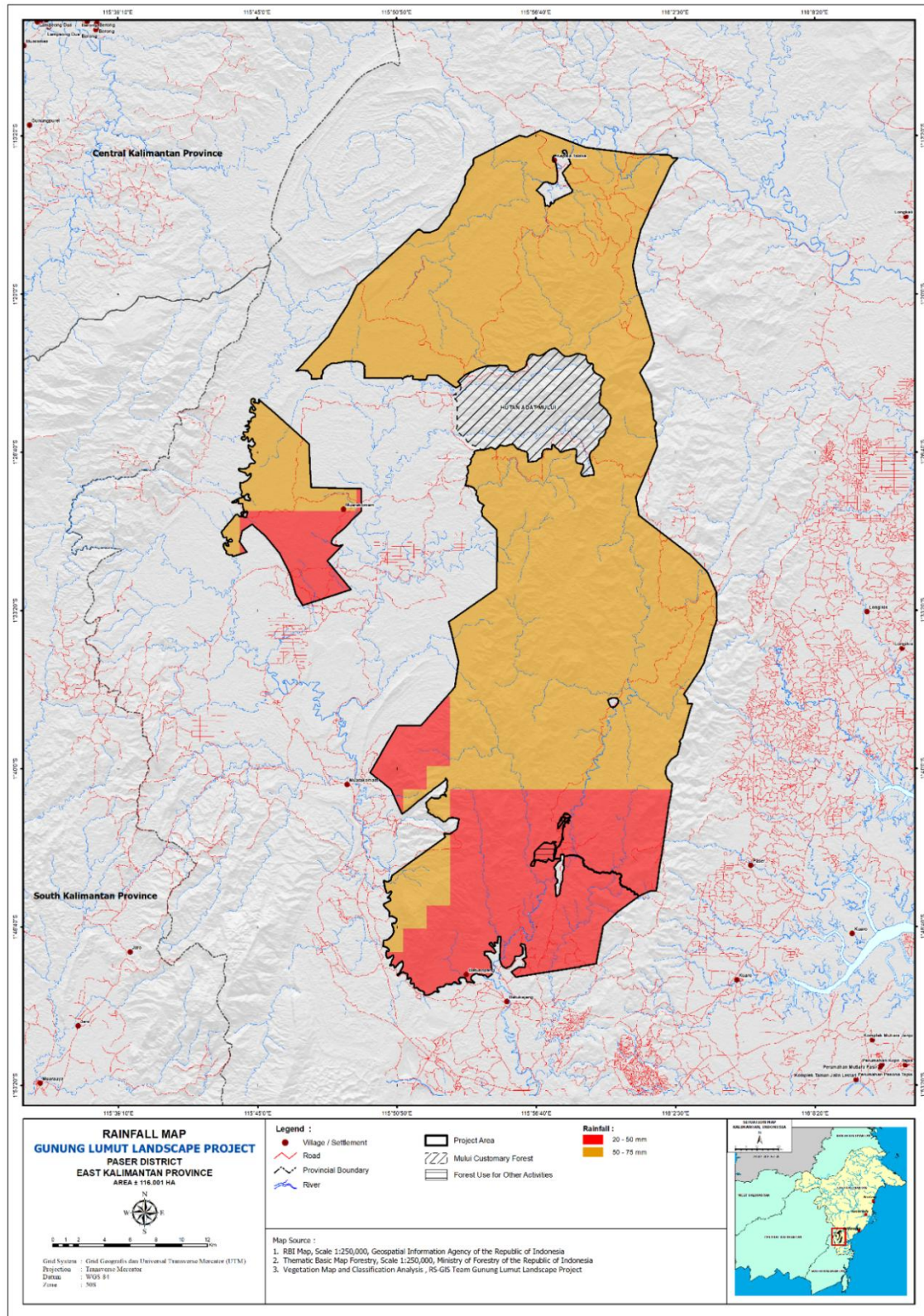


Figure 8. Rainfall Distribution Map in Project Area

5. Hydrology

Most of the project area falls within the Kendilo watershed. The area within Kendilo watersheds is approximately 79,594 Ha (68.62%) of the project area. Another watershed within the project area is the Tunan-Manggar watershed, with an area of 36,407 Ha or 31.38% of the project area (Refer to Table 6). Paser Regency has many rivers spreading across the region that also affects the hydrological conditions of Paser Regency. Five districts have river crossings spreading across village areas such as Muara Komam District with 30 rivers, Batu Sopang District with 29 rivers, Long Ikis District with 33 rivers, Long Kali District with 52 rivers, and Kuaro District with 50 rivers²⁶. Swamp areas can also be found in Kuaro, Long Ikis, and Long Kali districts (Table 6 and Figure 9).

Table 6. Watersheds in the Project Area

No	Watersheds	Area (Ha)	Percentage (%)
1	Kendilo	79,594	68.62
2	Tunan - Manggar	36,407	31.38
Total		116,001	100.00

²⁶ PT TMK High Conservation Value Identification Study Report, 2023, 20

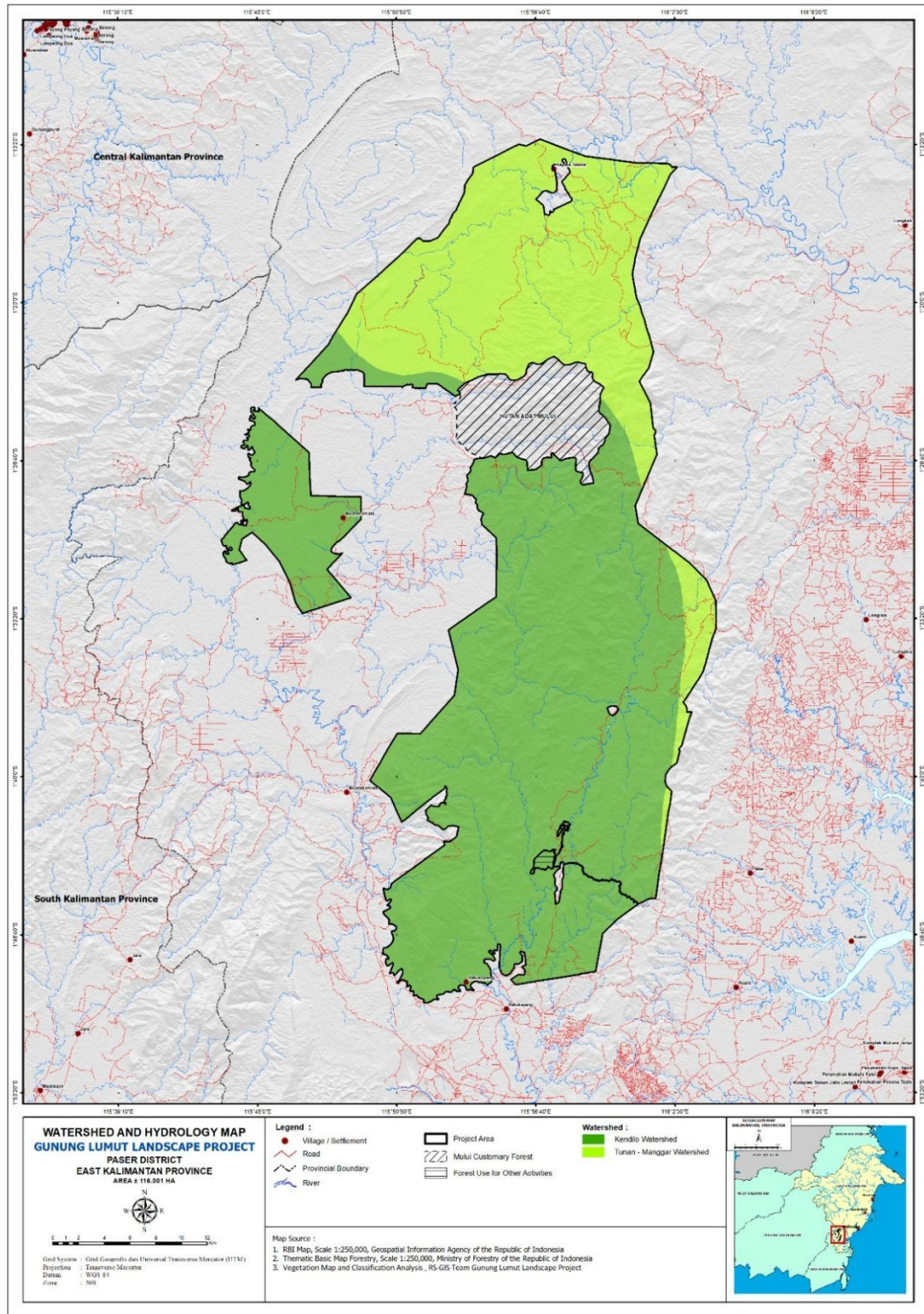


Figure 9. Map of Hydrology and Watershed in Project Area

6. Types of Vegetations

Based on the Periodic Forest Carbon Inventory (IKHB) results conducted in 2022, at least 188 identified species were found in the project area. Of the species found, the most common species were from the Dipterocarpaceae family (27 species), then successively from the Moraceae family (16 species), Leguminosae (13 species), Anacardiaceae (10 species) and Lauraceae (10 species)²⁷

The types of trees that mainly make up the tree stands in the project area, which have an IVI (Important Value Index) value of > 5% based on their growth rate, are as reflected in Table 7:

Table 7. Main Constituent TreeTypes²⁸

Seedling Level

Species	DT	RD (%)	F	RF (%)	IVI (%)
Shorea sp	2,364.00	11.81	0.30	10.14	21.94
Leea indica	1,439.00	7.19	0.16	5.53	12.72
Nephrolepis sp.	1,336.00	6.67	0.12	4.15	10.82
Calathea sp.	1,473.00	7.36	0.07	2.30	9.66
Aglaia simplicifolia	994.00	4.96	0.14	4.61	9.57
Peronema canescens	788.00	3.94	0.12	4.15	8.08
Selaginella sp.	1,131.00	5.65	0.03	0.92	6.57

Sapling Level

Species	DT	RD (%)	DM	DR (%)	F	RF (%)	IVI (%)
Macaranga hypoleuca	269.00	8.49	0.37	5.10	0.16	3.85	17.44
Peronema canescens	209.00	6.60	0.51	7.07	0.15	3.53	17.19
Shorea sp	165.00	5.21	0.29	3.95	0.18	4.17	13.32
Shorea macrobalanos	181.00	5.71	0.33	4.61	0.11	2.56	12.89
Aglaia simplicifolia	154.00	4.86	0.25	3.41	0.19	4.49	12.76
Baccaurea tetandra	83.00	2.62	0.27	3.75	0.12	2.88	9.25
Palaquium quercifolium	110.00	3.47	0.18	2.51	0.11	2.56	8.54
Polyalthia rumphii	72.00	2.27	0.17	2.30	0.12	2.88	7.45
Litsea sp.	88.00	2.78	0.17	2.35	0.10	2.24	7.37
Endertia spectabilis	66.00	2.08	0.15	2.05	0.11	2.56	6.70
Alseodaphne umbelliflora	55.00	1.74	0.19	2.65	0.10	2.24	6.63
Syzygium aqueum	61.00	1.93	0.21	2.94	0.07	1.60	6.47
Dillenia excelsa	50.00	1.58	0.17	2.32	0.10	2.24	6.14
Leea indica	83.00	2.62	0.11	1.49	0.08	1.92	6.04
Shorea ovata	61.00	1.93	0.14	1.88	0.08	1.92	5.73
Horsfieldia irya	55.00	1.74	0.17	2.30	0.07	1.60	5.64

²⁷ Main Report of Periodic Forest Carbon Inventory of PT TMK, 2022, IV-1

²⁸ Note: DT = Density, RD = Relative Density, F= Frequency, RF = Relative Frequency IVI = Important Value Index, DM = Dominance, DR = Relative Dominance

Species	DT	RD (%)	DM	DR (%)	F	RF (%)	IVI (%)
Mangifera sp.	39.00	1.23	0.12	1.67	0.10	2.24	5.14
Lumnitzera racemosa	39.00	1.23	0.14	1.92	0.08	1.92	5.07
Memecylon lilacinum	44.00	1.39	0.13	1.73	0.08	1.92	5.04

Pole Level

Species	DT	RD (%)	DM	DR (%)	F	RF (%)	IVI (%)
Peronema canescens	61.00	10.15	0.97	10.47	0.16	4.55	25.17
Macaranga hypoleuca	25.00	4.16	0.40	4.29	0.11	3.03	11.48
Gluta wallichii	24.00	3.99	0.41	4.39	0.10	2.65	11.03
Shorea macrobalanos	18.00	3.00	0.31	3.36	0.14	3.79	10.15
Macaranga gigantea	24.00	3.99	0.35	3.81	0.08	2.27	10.08
Syzygium aqueum	20.00	3.33	0.33	3.53	0.11	3.03	9.89
Pterospermum javanicum	16.00	2.66	0.27	2.89	0.07	1.89	7.45
Aglaia simplicifolia	14.00	2.33	0.21	2.32	0.08	2.27	6.93
Dillenia excelsa	11.00	1.83	0.20	2.20	0.05	1.52	5.54
Shorea sp	10.00	1.66	0.17	1.79	0.07	1.89	5.35
Shorea sumatrana	10.00	1.66	0.19	2.04	0.05	1.52	5.22
Litsea sp.	10.00	1.66	0.15	1.62	0.07	1.89	5.18
Hopea ferruginea	9.00	1.50	0.16	1.69	0.07	1.89	5.09

Small Tree Level

Species	DT	RD (%)	DM	DR (%)	F	RF (%)	IVI (%)
Peronema canescens	8.00	4.49	0.35	6.36	0.12	3.77	14.62
Macaranga sp.	6.00	3.37	0.26	4.67	0.10	2.93	10.97
Gmelina arborea	5.00	2.81	0.23	4.20	0.07	2.09	9.11
Endertia spectabilis	5.00	2.81	0.17	3.15	0.08	2.51	8.47
Eusideroxylon zwageri	4.00	2.25	0.18	3.18	0.08	2.51	7.94
Pterospermum javanicum	4.00	2.25	0.14	2.56	0.08	2.51	7.32
Aglaia simplicifolia	4.00	2.25	0.15	2.72	0.07	2.09	7.06
Shorea macrobalanos	3.00	1.69	0.14	2.55	0.08	2.51	6.74
Macaranga hypoleuca	3.00	1.69	0.14	2.50	0.07	2.09	6.28
Mangifera sp.	3.00	1.69	0.13	2.35	0.07	2.09	6.12
Glochidion sericeum	3.00	1.69	0.12	2.18	0.07	2.09	5.96
Shorea ovata	3.00	1.69	0.09	1.69	0.07	2.09	5.47

Large Tree Level

Species	DT	RD (%)	DM	DR (%)	F	RF (%)	IVI (%)
Shorea macrobalanos	3.00	1.69	0.85	3.00	0.30	3.50	8.18

Species	DT	RD (%)	DM	DR (%)	F	RF (%)	IVI (%)
Endertia spectabilis	4.00	2.25	0.96	3.38	0.18	2.07	7.70
Gmelina arborea	5.00	2.81	0.61	2.15	0.19	2.23	7.18
Shorea johorensis	3.00	1.69	0.71	2.52	0.23	2.70	6.90
Shorea sp	3.00	1.69	0.63	2.24	0.23	2.70	6.63
Alseodaphne umbelliflora	3.00	1.69	0.40	1.42	0.25	2.86	5.97
Peronema canescens	3.00	1.69	0.61	2.15	0.18	2.07	5.90
Shorea bracteolata	3.00	1.69	0.54	1.91	0.18	2.07	5.67

Of all the species found, 24 species were included in the IUCN (International Union for Conservation of Nature) Red List, as shown in the following Table 8:

Table 8. Flora Species Listed in The IUCN Red List

No	Latin Name	Family	IUCN Red List	Species Abundance (individual/ha)			
				Sapling	Pole	Small Tree	Large Tree
1	<i>Shorea laevis</i> Ridley	Dipterocarpaceae	VU	6			1
2	<i>Dryobalanops aromatica</i> C.F.Gaertn.	Dipterocarpaceae	VU				1
3	<i>Artocarpus anisophyllus</i> Miq.	Moraceae	VU	11	-	-	1
4	<i>Durio dulcis</i> Becc.	Malvaceae	VU	-	-	1	-
5	<i>Baccaurea lanceolata</i> (Miq.) Mull.Arg.	Phyllanthaceae	VU	11	3	1	-
6	<i>Garcinia celebica</i> Desr.	Clusiaceae	VU	6	3	-	-
7	<i>Hopea sangal</i>	Dipterocarpaceae	VU	206	-	-	-
8	<i>Shorea macrobalanos</i> P.S.Ashton	Dipterocarpaceae	VU	181	18	3	3
9	<i>Shorea rugosa</i> F.Heim	Dipterocarpaceae	VU	-	2	1	-
10	<i>Parashorea stellata</i> Kurz	Dipterocarpaceae	VU	50	5	2	3
11	<i>Mangifera torquenda</i> Kosterm.	Anacardiaceae	VU	-	-	-	1
12	<i>Durio kutejensis</i> (Hassk.) Becc.	Malvaceae	VU	11	3	1	1
13	<i>Shorea albida</i>	Dipterocarpaceae	VU	69	-	-	-
14	<i>Durio acutifolius</i> (Mast.) Kosterm.	Malvaceae	VU	11	3	2	1
15	<i>Eusideroxylon zwageri</i> Teijsm.&Binn.	Lauraceae	VU	28	6	4	2
16	<i>Dipterocarpus acutangulus</i>	Dipterocarpaceae	EN	274	-	-	-
17	<i>Shorea sumatrana</i> (Slooten) Desch	Dipterocarpaceae	EN	28	10	2	1
18	<i>Shorea pachyphylla</i> Ridl. ex Symington	Dipterocarpaceae	EN	11	-	1	-
19	<i>Shorea pauciflora</i> King	Dipterocarpaceae	EN	165	10	2	3

No	Latin Name	Family	IUCN Red List	Species Abundance (individual/ha)			
				Sapling	Pole	Small Tree	Large Tree
20	<i>Cinnamomum parthenoxylon</i> (Jack) Meisn.	Lauraceae	EN	-	-	-	1
21	<i>Shorea ovata</i> Dyer	Dipterocarpaceae	EN	61	6	3	2
22	<i>Dracontomelon costatum</i> Blume	Anacardiaceae	EN	-	-	-	1
23	<i>Dipterocarpus hasseltii</i> Blume	Dipterocarpaceae	EN	6	-	-	1
24	<i>Hopea ferruginea</i> Parijs	Dipterocarpaceae	CR	39	9	1	1

The data above shows that there is one species that fall into the "critically endangered (CR)" category, namely *Hopea ferruginea* Parijs from the Dipterocarpaceae family, where it has the species abundance of Saplings at 39 individuals/Ha, Poles at 9 individuals/ha, Small and Large Trees at each only one individual/ha.

2.1.15 Social Parameters (VCS, 3.18; CCB, G1.3)

This section is still under development

2.1.16 Project Zone Map and Project Location (VCS, 3.11, 3.18; CCB, G1.4-7, G1.13, CM1.2, B1.2)

The project zone area consists of 24 villages based on the administrative boundaries. The villages that are included in the project zone are listed in Table 9 below.

Table 9. Villages in the Project Zone

No	Village	District
1	Muara Kuaro	Muara Komam
2	Prayon	Muara Komam
3	Muara Payang	Muara Komam
4	Long Sayo	Muara Komam
5	Lusan	Muara Komam
6	Swan Slutung	Muara Komam
7	Batu Butok	Muara Komam
8	Uko	Muara Komam
9	Binangon	Muara Komam
10	Pinang Jatus	Long Ikis
11	Lombok	Long Ikis
12	Brewe	Long Ikis
13	Samuntai	Long Ikis
14	Long Gelang	Long Ikis
15	Tiwei	Long Ikis
16	Kepala Telake	Long Kali

No	Village	District
17	Muara Lambakan	Long Kali
18	Kasungai	Batu Sopang
19	Rantau Buta	Batu Sopang
20	Rantau Layung	Batu Sopang
21	Busui	Batu Sopang
22	Sungai Terik	Batu Sopang
23	Sandeley	Kuaro
24	Modang	Kuaro

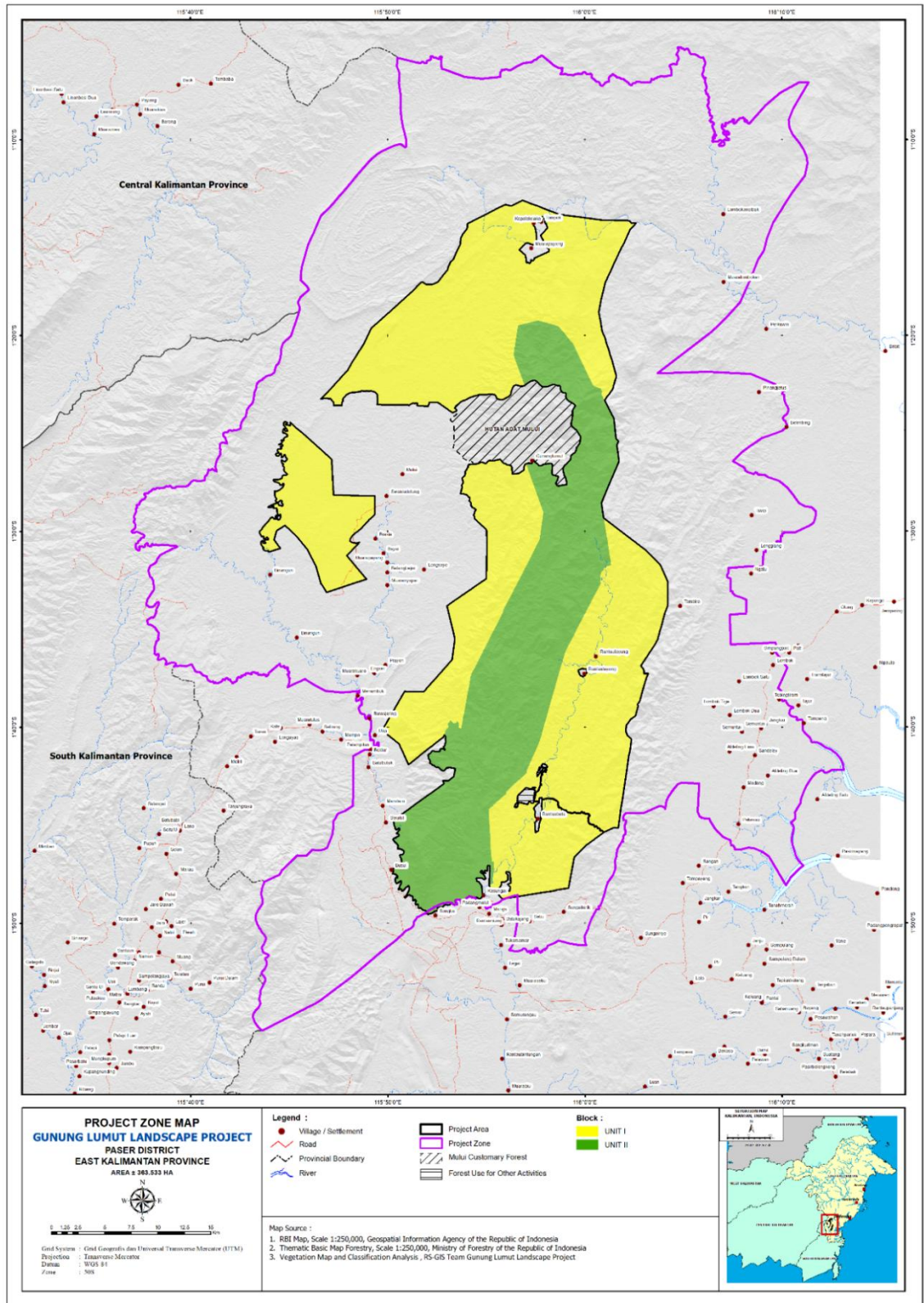


Figure 11. Map of Villages Within and Surrounding The Project Zone

2.1.17 Project Activities and Theory of Change (VCS, 3.6; CCB, G1.8)

GLLP aims to optimize the business permits given through multiple business schemes. One of the business schemes being developed is carbon project, where it will prevent GHG emissions by stopping logging activities, restoring land ecosystems, and improving the economic value that is currently in a degraded state caused by previous fire and logging activities. The activities are designed to maximize its capability to restore the project area's ecological functions and build relationships with stakeholders within the landscape. To achieve this, the project activities to be implemented are:

Climate and Environment Biodiversity

The project activities listed aim to avoid emissions and leakage from activities and events such as fires, deforestation, mining and illegal activities through activities such as land management and education on the issues caused by illegal activities. These project activities will also aim to protect various endangered species while providing new habitats and protecting the habitats of the wildlife through monitoring activities, education and establishing captive breeding.

Community

The project activities listed is committed to improve the economic, quality and the well-being of the people through activities such as development of community forest, utilizing natural medicinal plants, education, and NTFP alternative solutions. The project also considers sustainability goals by considering the customary roles, such as through strengthening customary roles within the community as well as traditional schools to enrich the knowledge of the younger generations in the sustainability of customary institutions.

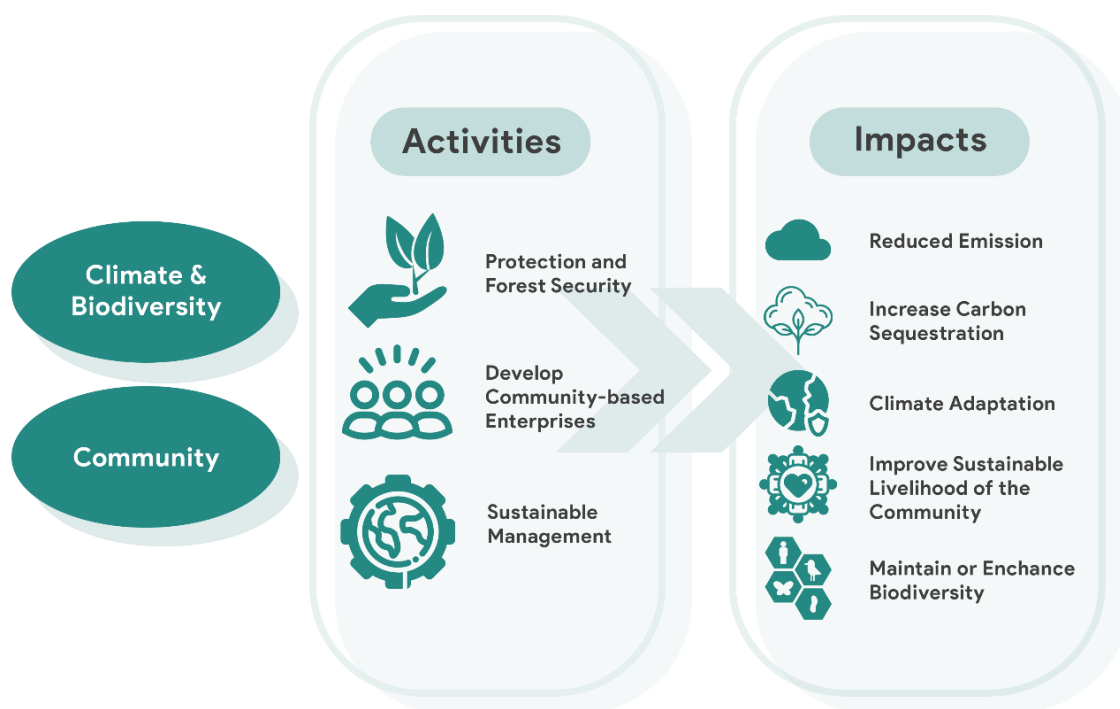


Figure 12. Diagram of Project Activities and Theory of Change

The GLLP project activities will also consider the aspects of theory of change, the activity description, expected climate, community, and/or biodiversity, and relevance of the activities to the project are explained in table 10 below.

Table 10. Theory of Change

Activity description	Expected climate, community, and/or biodiversity			Relevant to project activities
	Output (short-term)	Outcome (mid-term)	Impacts (long term)	
Forest Protection	Prevent forest degradation Decrease potential in fire outbreaks and illegal activities	Improve forest ecosystem	Avoid emission from fire and illegal activity, increasing removal	Climate, Biodiversity, Environment
Patrol and monitoring	Prevent forest degradation, prevent the illegal acts on the forest resources	Improve forest ecosystem	Avoid carbon emission removal, illegal activities reduced	Climate, Biodiversity, Environment

Activity description	Expected climate, community, and/or biodiversity			Relevant to project activities
	Output (short-term)	Outcome (mid-term)	Impacts (long term)	
Convert of harvested areas for protection	Increase number of protected areas	Avoiding forest degradation Provide an improve habitat for wildlife	Maintain or enhance biodiversity No source of emissions from legal harvesting	Climate, Biodiversity, Environment
Preventing conversions of forest into mining areas	No logging activity for mine land clearing	A fixed forest cover area	- No source of emissions from deforestation into mining land - Maintaining the forest function for ecological services - Protection of wildlife habitat and biodiversity	Climate, Biodiversity, Environment
Gender	Women Involvement in activities within the project zone	Increase participation of women's group	Increasing social roles and family income.	Community
Soil Management	Unsustainable farming system	Community empowerment to produce and use organic fertilizers, reduce environmental pollution, and sustainably improve land quality.	Long-term use of organic fertilizers can increase land productivity and prevent land degradation	Community
Assistance and community capacity building through community forest development	Community forests are optimally developed	- Provisions of wood needs - Alternative income	- Improvement of community welfare - Leakage emissions, which a lack of wood supply may have caused, did not occur - Provision of new habitats for wildlife	Community, Climate, Biodiversity, Environment

Activity description	Expected climate, community, and/or biodiversity			Relevant to project activities
	Output (short-term)	Outcome (mid-term)	Impacts (long term)	
Regional Public Company (Perseroda)	Expansion and improvement of the business scale of cooperatives.	The legitimacy and carrying capacity of the government will increase in society.	Permanent business due to regency government intervention.	Community
Customary roles and climate change	Strengthening the customary roles in the community life	Customary roles can regulate the use of forest products	The forest will be sustainable	Community
Customary education (sekolah adat)	Customary education will teach traditional customs to the younger generations	Increase the knowledge & skills of the younger generation as candidates for successors to traditional institutions.	Continuity of customary institutions	Community
Awareness and involvement of Illegal loggers, illegal mining	Transformation of mindset and illegal activities	Alternative income, Prevention of forest deforestation and degradation	Increasing community welfare Emission leaks that illegal activities may cause do not occur Protected habitat for wild animals and biodiversity	Community, Climate, Biodiversity Environment
Utilization and preservation of natural Medicinal Plants	One of the implementations of the Silvoforestry program	Cultivation of medicinal plants can increase people's income	Economic quality and public health will improve	Community
Provide good agricultural/NTF P best management practices training to local communities	Provide an alternative solution to gain more fix income for local communities	Establish agroforestry system for local communities and link with the market	Improve livelihood of the local communities, dependency reduction to illegal logging/hunting activities	Community, Biodiversity
Reforestation	Create job opportunities for local communities	Improve forest ecosystem	Increase carbon sequestration, habitat for wildlife	Climate, Biodiversity Environment and Community

Activity description	Expected climate, community, and/or biodiversity			Relevant to project activities
	Output (short-term)	Outcome (mid-term)	Impacts (long term)	
	and increase forest cover			
Increasing the role of social and economic institutions	Forming of cooperatives in each village	- An organized society with the capacity to be economically self-sufficient. - Communities have access to finance and capital - Communities have stronger bargaining positions in the market.	Cooperatives as legal entities have potential to develop along with the needs of the scales of business.	Community
Utilization of environmental services (carbon and other forest products)	Provide an alternative solution to gain more fix income for local communities	Establish agroforestry system for local communities and link with the market	Improve livelihood of the local communities, dependency reduction to illegal logging/hunting activities	Community, Biodiversity
Agroforestry Implementation	With the agroforestry system, the community has a legal livelihood pattern.	The community has a permanent livelihood without destroying nature.	People no longer carry out illegal activities as a livelihood.	Community, Climate and Biodiversity Environment
Cooperation and development of business entities	The need for cooperatives to develop business scale	- Cooperatives have the opportunity to cooperate with other legal entities to increase business scale by forming community-based businesses (social entrepreneurship) - Cooperatives have the opportunity to Collaborating with international certification bodies	- Forming a community-based company will increase the community's dignity as the owner of the company. - The high dignity of the community regarding their efforts will increase organizational	Community

Activity description	Expected climate, community, and/or biodiversity			Relevant to project activities
	Output (short-term)	Outcome (mid-term)	Impacts (long term)	
		to certify superior community products such as organic certification, fair trade certification, rain forest alliance certification and FSC certification	citizenship behavior. - Certification will increase the bargaining position of the product while increasing the selling price of the product.	
Development of natural resource cultivation outside the area (eg: captive breeding)	- Baseline utilization (taking) of various natural resources from the area (plants and wildlife) - Identified various options for sustainable use.	- Selected cultivation techniques (based on ease of technique, market access) developed - The community's cultivation capacity is built.	The community can still utilize natural resources in a sustainable manner without over-exploitation of natural resources	Community, Biodiversity

2.1.18 Sustainable Development Contributions (VCS, 3.17)

This project shall contribute to the achievement of sustainable development goals through the following:

SDG 1 – No Poverty

SDG 1 Goal can be achieved by:

- Improved livelihoods of local communities
- Creating new jobs for locals

Activities that will be carried out in the project to support this goal include: gender, soil management, assistance and community capacity building through community forest development, optimization of Regional Public Company (Perseroda), awareness building and involvement of Illegal loggers and illegal miners, utilization and preservation of natural medicinal plants, good agricultural/NTFP best management practices trainings to local communities, reforestation, increasing the role of social and economic institutions, utilization of environmental services (carbon and other forest products), agroforestry implementation, and cooperation and development of business entities.

SDG 5 – Gender Equality

SDG 5 Goal can be achieved by:

- Economic empowerment of women
- Inclusive and quality education

The empowerment of women's groups aims to increase their role and family income. Some of the performance of women's empowerment is by conducting coaching to rattan women reciters in Lusan, Rantau Buta & Layung Villages. Coaching micro business of making small food. In Prayon Village, several groups proposed the creation of hydroponic vegetable installations and PKK gardens as well as coaching for generating honey bees. Activities that will be carried out in the project to support this goal include: gender and customary education (sekolah adat).

At traditional events there is good cooperation among the community. Women play an important role in organizing traditional events. Women's groups voluntarily prepare the decorations and dishes for the guests. The men collect the food ingredients, including deer hunting, and then the women's group cooks the food ingredients into dishes to be served.

SDG 11 – Sustainable Cities and Communities

SDG 11 Goal can be achieved by:

- Regenerative approach to forest management
- Community participation in forest management

Activities that will be carried out in the project to support this goal include: Forest protection, patrol and monitoring, conversion of harvested areas for protection, prevention of conversion of forest into mining areas, soil management, assistance and community capacity building through community forest development, customary roles and climate change, customary education (sekolah adat), awareness building and involvement of illegal loggers and illegal miners, utilization and preservation of natural medicinal plants, good agricultural/NTFP best management practices training to local communities, reforestation, increasing the role of social and economic institutions, utilization of environmental services (carbon and other forest products), agroforestry implementation, development of natural resource cultivation outside the area (e.g.: captive breeding).

SDG 12 – Responsible Consumption and Production

SDG 12 Goal can be achieved by:

- Reducing forest deforestation and degradation
- Wood certification

Activities that will be carried out in the project to support this goal include : forest protection, patrol and monitoring, conversion of harvested areas for protection, preventing conversion of forest into mining areas, assistance and community capacity building through community forest development, awareness building and involvement of illegal loggers and illegal miners, good agricultural/NTFP

best management practices training to local communities, reforestation, and utilization of environmental services (carbon and other forest products).

SDG 13 – Climate Action

SDG 13 Goal can be achieved by:

- Carbon sequestration
- Climate change mitigation

Activities that will be carried out in the project to support this goal include : forest protection, conversion of harvested areas for protection, preventing conversion of forest into mining areas, assistance and community capacity building through community forest development, customary education (sekolah adat), good agricultural/NTFP best management practices training to local communities, reforestation, utilization of environmental services (carbon and other forest products).

SDG 15 – Life on Land

SDG 15 Goal can be achieved by:

- Restoring degraded forest to its natural state
- Increasing the value of biodiversity by connecting protected areas with concession areas

Activities that will be carried out in the project to support this goal include : conversion of harvested areas for protection, preventing conversion of forest into mining areas, and reforestation.

Infographic that showed the linkages between each activity and each sustainable development goal can be seen in Figure 13.

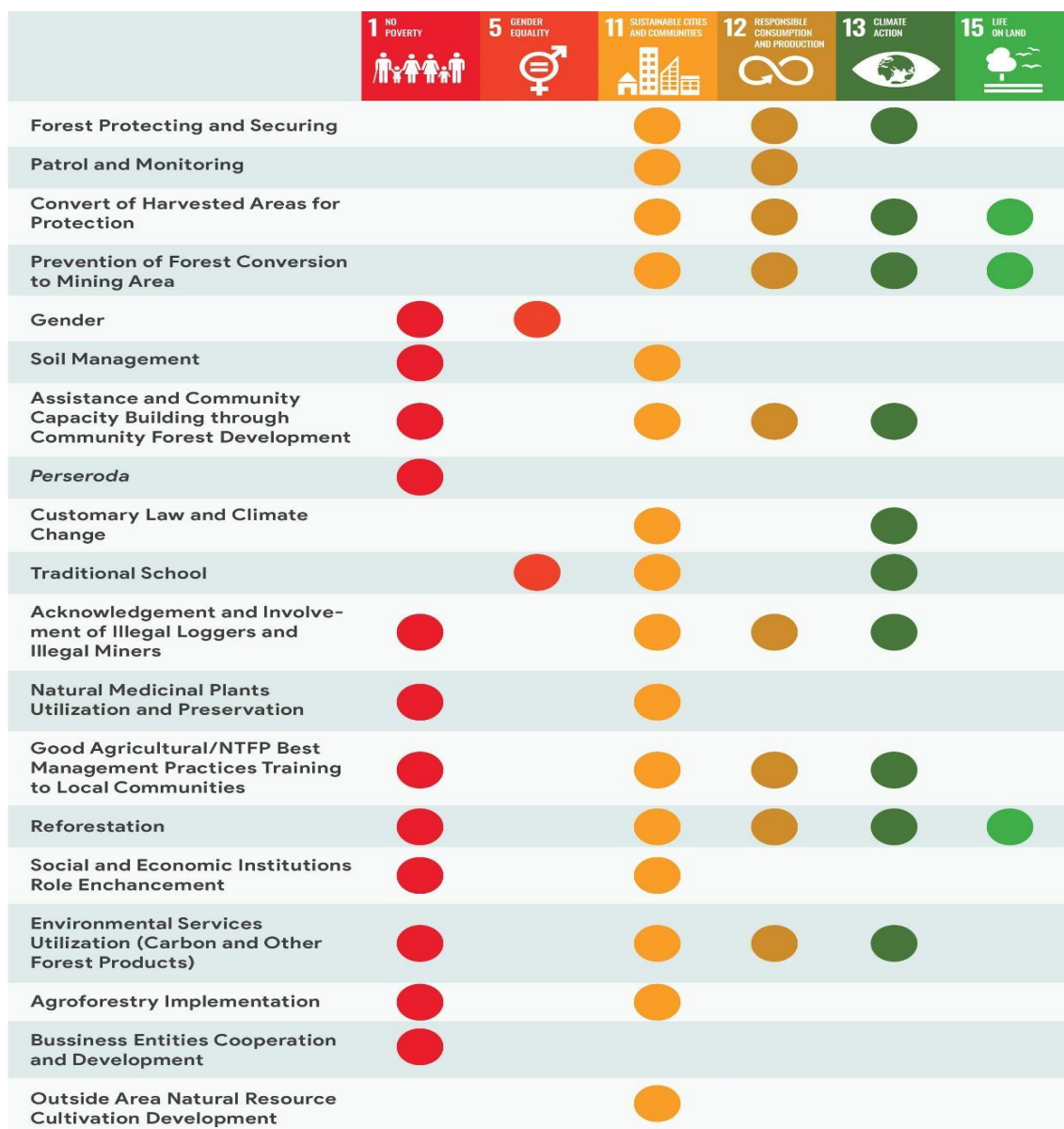


Figure 13. Infographic of Project Activities Relating to Sustainable Development Goals

2.1.19 Implementation Schedule (CCB, G1.9)

Date	Milestone(s) in the project's development and implementation
October 6, 2020	Carbon Feasibility Study Agreement with Hatfield
April 2021	Carbon Feasibility study of PT TMK
May 2021	Meeting with one of the coal mining companies in the country that started illegal mining activities in the concession area ("PT X") and then reporting this

Date	Milestone(s) in the project's development and implementation
	illegal activity to the Ministry of Energy and Minerals Resources as rejection of any illegal mining activities in the concession area
September 16, 2021	License Issuance: Decree of the Minister of Environment and Forestry Number SK SK.782/Menlhk/Setjen/HPL.0/9/2021 concerning PT TMK's Forest Utilization Permit Decree (PBPH) covering an area of 82,805 Ha, which provides PT TMK permission to manage the forest in the project area
March 22, 2022	Issuance of SK. 546/1/IUP/PMDN/2022 by the Ministry of Energy and Mineral Resources which granted mining business permit to PT X
September 8, 2022	Issuance of Warning Letter from the Ministry of Environment and Forestry Number S.745/PUPH/EKU/HPI.1/9/2022 regarding PT TMK's PBPH-HA Performance in East Kalimantan Province which requires PT TMK to realize a forest product production plan of at least 50% from the planned harvesting target
September 30, 2022	Decree of the Minister of Environment and Forestry Number SK.7631/MenLHK-PHL/PUPH/HPL.1/9/2022 concerning Approval of Work Plans for Forest Utilization Business Permits for Forest Utilization Period 2022-2031 on behalf of PT TMK
December 2022	Received information that PT X is submitting an application for Approval for the Use of Forest Areas ("PPKH") to the Ministry of Environment and Forestry to carry out mining activities in the project area
January 13, 2023	Letter No. 522.1/323/DK-II/2022 regarding Technical Considerations for PT TMK's PBPH Application for HL Gunung Lumut was issued by the Head of the East Kalimantan Provincial Forestry Service
February 10, 2023	Letter No. 503/0022/PBPH/DPMPTSP/II/2023 regarding PBPH Recommendations for HL Gunung Lumut was issued by the Head of the Investment and One-Stop Services Agency (DPMPTSP) on behalf of the Governor of East Kalimantan Province to PT TMK
February 14, 2023	Request for Revision of RKUPH ("RKU MUK") submitted to the Director General of PHL, with the aim of eliminating logging obligations for PT TMK on and over the project area
February 15, 2023	The PBPH application for HL Gunung Lumut is submitted to the Minister of KLHK c.q. the Head of OSS
February 20 2023	PT TMK was selected as a pioneer in the Multi Forestry Business ("MUK") pilot project by the Indonesian Chamber of Commerce and Industry ("KADIN")

Date	Milestone(s) in the project's development and implementation
February 22, 2023	Letter of Request for Support for the Implementation of the MUK Pilot Project at PT TMK to the Director General of PHL, which was submitted in relation to the certainty and security of the business area, which was triggered by the application of PPKH by PT X
February 27, 2023	Request for Support for the Implementation of the MUK Pilot Project at PT TMK to the Secretary General of the Ministry of Environment and Forestry, which was submitted in relation to the certainty and security of the business area, which was triggered by the application for PPKH submitted by PT X
March 8, 2023	Request for Support for the Implementation of the MUK Pilot Project at PT TMK to the Secretary General of the Ministry of Environment and Forestry, which was submitted in relation to the certainty and security of the business area, which was triggered by the application for PPKH submitted by PT X
March 15, 2023	Stipulation of the decree from the Ministry of Environment and Forestry for PPKH PT X on the operational activities of coal mining and its supporting facilities.
February 20 2023 – March 5 2023	Participatory Rural Appraisal (PRA)
March 28, 2023	Information on the RKU MUK submission process to the Director General of PHL so that the implementation of PT TMK's obligations under the RKUPH can be postponed until the RKU MUK is approved
February 20 – April 5, 2023	Biodiversity Monitoring
May 7 – 20, 2023	Stump Survey
November 2, 2023	The project's long-term Forest Utilization Work Plan (RKU) for PT TMK has been formally ratified through a Ministerial Decree issued by the Ministry of Environment and Forestry of the Republic of Indonesia (e.g., SK.782/MENLHK/SETJEN/HPL.0/9/2021), which confirms the approval and revision of the RKU for the 2022–2031 period. Number: SK.11658/MenLHK-PHL/PUPH/HPL.1/11/2023

2.1.20 Risks to the Project (CCB, G1.10)

This section is still under development

2.1.21 Benefit Permanence (CCB, G1.11)

This section is still under development

2.1.22 Financial Sustainability (CCB, G1.12)

This section is still under development

2.2 Without-project Land Use Scenario and Additionality

2.2.1 Conditions Prior to Project Initiation and Land Use Scenarios without the Project (VCS, 3.13; CCB, G2.1)

Before the GLLP, the project area was designated for production forest and protected forest area. The production forest area (Unit I) has been logged since the mid-1990s. As a result, it is a complex mosaic forest area with full until very degraded state (see figure 14) The supposed land use without project scenario that is most possible for Unit I area is forest management activities for logging/wood production. With the project, this area will be converted from a production forest area to a protected area. This shows that the project will not generate GHG emissions, yet rather it will reduce or further eliminate emissions. In addition, according to the applicable laws and regulations, the Ministry of Environment and Forestry (“KLHK”) has the right to grant lease-to-use forest area (“PPKH”) permits for legal mining companies to conduct operational mining activities up to 10% of the total forest area. The PPKH issuance triggers legal mining operation activities. However, it is also known that the people within and surrounding the project area have also carried out illegal mining activities within the project area. With the GLLP project activities, a review on and further rejection proposal to the submission of PPKH proposals can be submitted to KLHK, as well as prevention measures regarding illegal mining activities can be implemented, therefore mining activities in the project area can be monitored and reduced in intensity.



Figure 14. Various Forest Cover Condition in GLLP

For the protected forest area (Unit II) without the project, the management of the protected forest was previously carried out by the provincial government, which then had a policy change. Currently, it is carried out by a site-level management unit known as the Forest Management Unit (“KPH”). The management of the Gunung Lumut protected forest is included in the management area of KPH Telake. In the Long Term Forest Management Plan (RPJHP), the protected forest area of

Gunung Lumut is included in the utilization zone. Without the project, Gunung Lumut's protected forest is only managed as a preservation area without the function of utilizing environmental services that could allow development to optimize area utilization as stipulated in Government Regulation of the Republic of Indonesia No. 23 of 2021 regarding Forestry Management. Currently, management constraints are occurring so that it is not optimally managed. KPH Telake has a vast management area with a limited budget and human resources.

This causes the monitoring, protection, and security of the area, especially for Unit II to be very vulnerable. Meanwhile, in terms of benefits the limited budget and human resources resulted in less optimal management making less impact from an economic and social perspective²⁹. This and other possible scenarios are further discussed in sections 3.

2.2.2 Most-Likely Scenario Justification (CCB, G2.1)

This section is still under development

2.2.3 Community and Biodiversity Additionality (CCB, G2.2)

This section is still under development

2.2.4 Benefits to be used as Offsets (CCB, G2.2)

This section is still under development

2.3 Safeguards and Stakeholder Engagement

2.3.1 Stakeholder Identification (VCS, 3.18, 3.19; CCB G1.5)

This section is still under development

2.3.2 Stakeholder Descriptions (VCS, 3.18, 3.19; CCB, G1.6, G1.13)

This section is still under development

2.3.3 Stakeholder Access to Project Documents (VCS, 3.18, 3.19; CCB, G3.1)

This section is still under development

2.3.4 Dissemination of Summary Project Documents (VCS, 3.18, 3.19; CCB, G3.1)

This section is still under development

2.3.5 Informational Meetings with Stakeholders (VCS, 3.18, 3.19; CCB, G3.1)

This section is still under development

²⁹ Rencana Jangka Panjang Pengelolaan Hutan KPHP Telake

2.3.6 Risks from the Project and No Net Harm (VCS, 3.18, 3.19)

This section is still under development

2.3.7 Community Costs, Risks, and Benefits (CCB, G3.2)

This section is still under development

2.3.8 Information to Stakeholders on Validation and Verification Process (VCS, 3.18.6, 3.19; CCB, G3.3)

This section is still under development

2.3.9 Stakeholder Consultations (VCS, 3.18; CCB, G3.4)

This section is still under development

2.3.10 Continued Consultation and Adaptive Management (VCS, 3.18; CCB, G3.4)

This section is still under development

2.3.11 Stakeholder Consultation Channels (CCB, G3.5)

This section is still under development

2.3.12 Stakeholder Participation in Decision-Making and Implementation (VCS, 3.18, 3.19; CCB, G3.6)

This section is still under development

2.3.13 Anti-Discrimination Assurance (VCS 3.19; CCB, G3.7)

This section is still under development

2.3.14 Feedback and Grievance Redress Procedure (VCS, 3.18.4; CCB, G3.8)

This section is still under development

2.3.15 Accessibility of the Feedback and Grievance Redress Procedure (VCS, 3.19; CCB, G3.8)

This section is still under development

2.3.16 Worker Training (VCS, 3.19; CCB, G3.9)

This section is still under development

2.3.17 Community Employment Opportunities (VCS, 3.19.13; CCB, G3.10)

This section is still under development

2.3.18 Occupational Safety Assessment (VCS, 3.19; CCB, G3.12)

This section is still under development

2.4 Management Capacity

2.4.1 Project Governance Structures (CCB, G4.1)

This section is still under development

2.4.2 Required Technical Skills (VCS, 3.19; CCB, G4.2)

This section is still under development

2.4.3 Management Team Experience (VCS, 3.19; CCB, G4.2)

This section is still under development

2.4.4 Project Management Partnerships and Team Development (VCS, 3.19; CCB, G4.2)

This section is still under development

2.4.5 Financial Health of Implementing Organization(s) (CCB, G4.3)

This section is still under development

2.4.6 Avoidance of Corruption and Other Unethical Behavior (VCS, 3.19; CCB, G4.3)

This section is still under development

2.4.7 Commercially Sensitive Information (VCS, 3.5.2 – 3.5.4; CCB Rules, 3.5.13 – 3.5.14)

This section is still under development

2.5 Legal Status and Property Rights

2.5.1 National and Local Laws (VCS, 3.1, 3.6, 3.7, 3.14, 3.18, 3.19; CCB, G5.6)

GLLP is designed and implemented in full compliance with both provincial and national law. Relevant laws and regulations on land use, forestry, and climate and its relevance to the project are included and explained in the table 13 below:

Table 13. National and Local Laws

Regulation Name	Relevance to the Project
Law Number 1 of 1970 on Worker Safety Rules	This regulation stipulates that everyone who is in the workplace is guaranteed safety and that every source of production needs to be used and used safely and efficiently.
Law Number 7 of 1984 on Convention On The Elimination Of All Forms Of Discrimination Against Women	This regulation regulates the prohibition of all forms of discrimination against women
Law Number 5 of 1990 on Conserving Natural Resources and Its Ecosystem	This regulation aims to realize the preservation of living natural resources and the balance of their ecosystems so that they can better support efforts to improve people's welfare and the quality of human life.
Law Number 21 of 1999 ratifies International Labor Organization (ILO) No. 111 concerning Discrimination in Respect of Employment and Occupation	This regulation regulates the prohibition of all forms of discrimination against workers based on race, skin color, gender, religion, political views, nationality or heredity.
Law Number 19 of 2004 on Stipulation of Government Regulation in Lieu of Law Number 1 of 2004 on Amendment to Law No. 41/1999 on Forestry	This regulation regulates legal uncertainty in doing business in the mining sector in forest areas because of Law Number 41 of 1999 concerning Forestry does not regulate the continuity of permits or agreements existing mines.
Law Number 41 of 1999 on Forestry	This regulation is to optimize the various functions of forests which include conservation functions, protection functions, and production functions to achieve balanced and sustainable environmental, social, cultural and economic benefits.
Law Number 32 Year 2009 on Environmental Protection and Management; October 3, 2009	The purpose of this Law is to create an environmentally sustainable development through means of an environmental planning policy, and the rational exploitation, development, maintenance, restoration, supervision and control of the environment.
Law Number 18 of 2013 on The Prevention and Eradication of Forest Destruction	This regulation regulates the Prevention and Eradication of Forest Destruction

Regulation Name	Relevance to the Project
	by communities, legal entities, and/or corporations that obtain forest utilization permits, causing damage to sociocultural life and the environment and increasing global warming.
Law Number 16 of 2016 on Ratification of the Paris Agreement on the United Nations Framework Convention on Climate Change	This regulation is intended to recognize national commitments in reducing emissions from various sectors, conserving forests, increasing renewable energy and the role of local communities and indigenous peoples in controlling climate change.
Law Number 9 of 2018 on Non-Tax State Revenues	Utilization of natural resources is included in the types of objects from Non-Tax State Revenue (PNBP) because it obtains direct or indirect benefits from services or utilization of resources and rights obtained by the state.
Government Regulation Number 60 of 2009 on Concerning Amendment To Government Regulation Number 45 Of 2004 Concerning Forest Protection	This regulation regulates forest protection and nature conservation which aims to protect forests, forest areas and their environment, so that the functions of protection, conservation, and production are optimally and sustainably achieved.
Government Regulation Number 50 of 2012 on Occupational Health and Safety Management System	This regulation regulates all activities to guarantee and protect the safety and health of workers through efforts to prevent work accidents and work-related diseases.
Government Regulation Number 12 of 2014 on Concerning Type And Rates Of Non-Tax State Revenue Applicable To The Ministry Of Forestry	This regulation regulates Tariffs on Types of Non-Tax State Revenues that apply to the Ministry of Environment and Forestry
Government Regulation Number 5 of 2021 on Implementation Of Risk-Based Business Licensing	Business Licensing is the legality granted to Business Actors to start and run their business and/or activities based on the Risk level of business activities
Government Regulation Number 21 of 2021 on the Implementation of Spatial Planning; 02 February 2021	New regulation as set out in the Omnibus Law to allow clearer and streamlining of Spatial Planning. PT

Regulation Name	Relevance to the Project
	TMK Project need to follow this regulation for its spatial plan.
Government Regulation Number 22 of 2021 on Environmental Protection, Organisation and Management	This regulation regulates systematic and integrated efforts made to preserve environmental functions and prevent environmental pollution and/or damage which includes planning, utilization, control, maintenance, supervision, and law enforcement.
Government Regulation Number 23 of 2021 on the Implementation of Forestry; 02 February 2021	New regulation as set out in the Omnibus Law to regulate the implementation forestry practices in PT TMK Project. New multibusiness scheme in forestry practices is first introduced in this regulation
Government Regulation Number 24 of 2021 Procedure for Imposing Administrative Sanctions and Procedures for Non-Tax State Revenues Derived from Administrative Fines in the Forestry Sector 02 February 2021	PNBP is a levy paid by an individual or entity by obtaining direct or indirect benefits from services or utilization of resources and rights obtained by the state
Government Regulation No 35 of 2021 on Fixed Term Employment Contract, Outsourcing, Working Time and Rest Time, and Termination	This law regulates: 1) Specific Time Work Agreement (PKWT) based on a certain period of time or completion of work; 2) type and nature or activity of work, period of time, and deadline for PKWT extension; 3) compensation money for PKWT Workers/Labourers; 4) worker/labourer protection and business licensing for outsourcing activities; 5) working time in certain business sectors or jobs; 6) Overtime Work Time and Overtime Wage; 7) certain Company restrictions that can apply long breaks; 8) procedures for Termination of Employment; and 9) provision of severance pay, gratuity pay, and rights replacement pay.
Government Regulation No 36 of 2021 concerning Wages	This law regulates wage policy, setting wages based on time units and/or output units, wage structure and scale, minimum wages, lowest wages for micro and small businesses, wage protection, form and method of payment of wages, matters that can be

Regulation Name	Relevance to the Project
	calculated by wages, wages as the basis for calculating or paying other rights and obligations, wage councils, and administrative sanctions. Wages consist of components: 1) wages without allowances; 2) fixed basic wages and allowances; 3) basic wages, fixed allowances and non-fixed allowances; or 4) base wages and allowances are not fixed.
Government Regulation No 37 of 2021 concerning The Implementation of the Job Loss Security Program	This law regulates participation, contributions, benefits, implementation of a Job Loss Guarantee program (JKP) that is integrated in an effective system, sources of funding, and administrative sanctions. JKP is social security provided to workers/laborers who experience termination of employment in the form of cash benefits, access to job market information, and job training. Entrepreneurs are required to include Workers/Laborers as participants in the JKP program. The JKP program is held to maintain a decent standard of living when workers/laborers lose their jobs. JKP is organized by BPJS Labor and central government
Government Regulation in Lieu of Law Number 2 of 2022 on Job Creation	This Government Regulation in Lieu of Law on Job Creation includes: 1) enhancing the investment ecosystem and business activities; 2) employment; 3) convenience, protection, and empowerment of Cooperatives and UMK-M; 4) ease of doing business; 5) research and innovation support; 6) land acquisition; 7) economic area; 8) Central Government investment and acceleration of national strategic projects; 9) implementation of government administration; and 10) imposition of sanctions.
Regulation of the President of The Republic of Indonesia Number 98 of 2021 on The Implementation of Carbon Pricing to Achieve the	This regulation regulates the economic value of carbon as one of the instruments in realizing the

Regulation Name	Relevance to the Project
Nationally Determined Contribution Target and Control over Greenhouse Gas Emissions in the National Development	Government's obligation to contribute to reducing greenhouse gas emissions, through selecting the most efficient, effective and just mitigation and adaptation actions without reducing the contribution targets set nationally.
Presidential Decree Number 32 of 1990 on Management of Protected Areas; July 25, 1990	The objective of this Presidential Regulation is to prevent damage to the environmental sector. The goal is managing the protection of the nature, climate, plants and animals and the cultural heritage as well as the historical and cultural values of the nation. It seeks to maintain plant diversity, animals, system types, and natural uniqueness. PT TMK Project use the regulation as the main principle for protection of environment and its biodiversity
Presidential Instruction Number 3 of 2020 on Forestry and Land Fire Management	This regulation regulates efforts to control forest and land fires, which include activities for preventing, extinguishing, handling after forest and land fires as well as law enforcement efforts against criminal acts of forest and land fires.
Presidential Regulation of the Republic of Indonesia Number 110 of 2025 Concerning The Implementation of Carbon Economic Value Instruments and National Greenhouse Gas Emission Control	Provides the national policy framework that legitimizes and monetizes GLLP's AFOLU carbon offsets by aligning them with Indonesia's carbon economic value instruments and national GHG emission control.
Regulation of the Ministry of Forestry of the Republic of Indonesia Number P. 19/Menhut-II/2011 on Boundary Arrangement of Forest Utilization Permits Working Area	Holders of IUPHHK permits are required to carry out delineation of working area boundaries no later than 1 (one) year after the Decree is issued by the Ministry in accordance with the stages specified in the regulations.
Regulation of the Ministry of Forestry of the Republic of Indonesia Number P.43/Menhut-II/2013 on Boundary Arrangement of Forest Utilization Permits Work Area, Principle Approval of Forest Area Use, Principle Approval of Release of Forest Areas and	Holders of IUPHHK permits are required to carry out delineation of working area boundaries no later than 1 (one) year after the Decree is issued by the Ministry in accordance with the stages specified in the regulations. The

Regulation Name	Relevance to the Project
Management of Forest Areas in Forest Management Units and Forest Areas with Special Purpose	boundaries of the working area include the boundaries of the company and the partnership.
Ministry of Forestry Regulation No. P.32 / MenLHK / Setjen / Kum.1 / 3/2016 on Forest and Land Fire Control; 15 March 2015	This regulation stipulates general guidelines for managing forest and/or land fires, including organizing businesses/activities/actions, managing human resources and infrastructure as well as operational prevention, suppression, post-fire handling, evacuation and rescue support, and management support for forest fire control and and/or land
Ministry of Environment and Forestry Regulation Number P.71/Menlhk/Setjen/Kum.1/12/2017 on Implementation of the National Registry System for Climate Change Control; December 29, 2017	The National Registry System for Climate Change Control is a system for managing and providing web-based data and information about actions and resources for Climate Change Adaptation and Mitigation in Indonesia which is carried out with various resource supports;
Ministry of Environment and Forestry Regulation Number P.72/Menlhk/Setjen/Kum.1/12/2017 on Guidelines for Implementation of Measurement, Reporting and Verification and Resources for Climate Change Control; December 29, 2017	Measurement, reporting and verification of climate change control actions and resources need to be carried out in order to ensure accountability for the achievements of actions and resources in an accurate, transparent and accountable manner
Ministry of Environment and Forestry Regulation P.73/Menlhk/Setjen/Kum.1/12/2017 on Guidelines for Implementation and Reporting of National Greenhouse Gas Inventory; December 29, 2017	The GHG Inventory Implementation and Reporting Guidelines are intended to provide a reference in the implementation of GHG emission inventories at the national, provincial and/or regency/city levels so that they can be trusted, accurate, consistent.
Regulation of the Ministry of Environment and Forestry of the Republic of Indonesia Number P.8/MENLHK/SETJEN/KUM.1/3/2018 on Fixed Procedures for Field Checking Information on Hotspots and / or Information on Forest and Land Fires	Regulations serve as guidelines for management units to check hotspots and/or forest and land fire information.
Regulation of the Ministry of Environment and Forestry of the Republic of Indonesia Number P.9/MENLHK/SETJEN/KUM.1/3/2018 on Technical	Regulations on the alert status or forest and land fire emergency status in order to improve coordination,

Regulation Name	Relevance to the Project
Criteria for Forest and Land Fire Preparedness and Emergency Status	synchronization and integration as well as quick response to efforts to handle forest and/or land fires early
Regulation of the Ministry of Environment and Forestry Number P.106/Menlhk/Setjen/Kum.1/12/2018 on the Second Amendment to the Regulation of the Ministry of Environment and Forestry Number P.20/Menlhk/Setjen/Kum.1/6/2018 on Types of Protected Plants and Animals	This regulation stipulates protected plant species and/or changes from protected plants to non-protected plants, taking into account the preservation of living natural resources by utilizing living natural resources and the factual conditions of plant populations in nature and in society.
The Regulation of the Ministry of Environment and Forestry Number P.20/Menlhk/Setjen/Kum.1/6/2018 on Types of Protected Plants and Animals	This regulation list and stipulates the protected animal and plant species in Indonesia
The Regulation of the Ministry of Environment and Forestry Number P.92/Menlhk/Setjen/Kum.1/8/2018 on Regarding Amendments to the Regulation of the Minister of Environment and Forestry Number P.20/Menlhk/Setjen/Kum.1/6/2018 Concerning Protected Plants and Animals	This regulation list and stipulates the protected animal and plant species in Indonesia
The Regulation of the Ministry of Environment and Forestry Number 3 of 2021 Concerning Standards for Business Activities in the Implementation of Risk-Based Business Licensing in the Environment and Forestry Sector	This regulation regulates the Standards for Business Activities in the Implementation of Risk-Based Business Licensing in the Environment and Forestry Sector.
Regulation of the Ministry of Environment and Forestry Number 7 of 2021 concerning Forestry Planning, Changes in the Designation of Forest Areas and Changes in the Functions of Forest Areas, as well as the Use of Forest Areas	This regulation regulates the Delimitation of Working Areas for Forest Utilization Permits with the aim of providing legal certainty regarding the status, location, boundaries and extent of work.
Regulation of Ministry of Environment and Forestry Number 8 of 2021 on the Forestry Management and the Preparation of Forestry Management Plan, as well as Forestry Utilization in Protected Forest and Production Forest	This regulation regulates Forest Management and Preparation of Forest Management Plans, as well as Forest Utilization in Production Forests
Regulation of Ministry of Environment and Forestry Number 9 of 2021 on Management of Social Forestry	This regulation provides guidelines in granting management rights, permits, partnerships and Customary Forests in the field of Social Forestry. The guidelines also resolve issues of tenure

Regulation Name	Relevance to the Project
	and justice for local communities and customary law communities living in or around forest areas in the context of community welfare and preservation of social functions.
Regulation Ministry of Environment and Forestry Number 11 of 2022 on the Profession and Competence of Forest Management Technical Personnel	This regulation stipulates that every permit holder seeking to utilize production forests and use protected forests must have and/or employ technical forest management personnel
Regulation of Ministry of Environment and Forestry Number 21 of 2022 on the Guidelines of Carbon Economic Value Implementation	This regulation regulates the achievement of nationally determined contribution targets and control of greenhouse gas emissions
Regulation of Ministry of Environment and Forestry Number 7 of 2023 Concerning Forestry Sector Carbon Trading Procedures	This regulation regulates the Procedures for Carbon Trading in the Forestry Sector.
Decree of the Ministry of Environment and Forestry of the Republic of Indonesia Number SK.716/MENLHK/SETJEN/KUM.1/6/2023 Concerning the Integration of Carbon Economic Value Mechanisms in Agreements, Partnerships and Business Permits in the Environment and Forestry Sector	This decree stipulates that Partnership Agreements and Business Permits in the Environment and Forestry Sector must be integrated with climate change protection to achieve the nationally determined target of reducing greenhouse gas (GHG) emissions (NDC).
Regulation of The Ministry of Labor Number 5 of 2018 Concerning Occupational Health and Safety Work Environment	This regulation regulates all activities to guarantee and protect the safety and health of workers through efforts to prevent work accidents and work-related diseases
Regulation of The Ministry of Labor Number 5 of 2021 on Implementation of Assurance Program Namely in Work-related Accidents, Death and Pension Benefits	This regulation regulates the Procedures for Implementing the Work Accident Benefit, Death Benefit and Old Age Benefit Programs for employees involved in project activities
Regulation of the Director General of Forestry Planning Number P.5/VII-KUH/2011 on Technical Guidelines for Implementation of Boundary Arrangement of Forest Utilization Permits	This regulation serves as a guideline and reference in structuring the boundaries of the IUPHHK working area so that the boundaries, area and location of the IUPHHK working area are realized.

Regulation Name	Relevance to the Project
Regulation of the Director General of Social Forestry and Environmental Partnership Number P.18/PSKL/SET/PSL.0/11/2016 on Guidelines for Preparation of Cooperation Agreement Drafts	Regulations serve as guidelines for the preparation of a Cooperation Agreement (NKK) in a transparent, participatory, accountable and non-discriminatory manner by providing equal opportunities for men and women
Regulation of the Director General of Sustainable Production Forest Management Number P.5/PHPL/UHP/PHPL.1/2/2016 on Guidelines for Potential Mapping and Conflict Resolution in Holders of Business Permit for Utilization of Timber Forest Products (IUPHHK) in Production Forests	Regulations serve as guidelines and references for the management unit (IUPHHK) in mapping potential conflicts and resolution of conflicts to identify potential conflicts and to formulate efforts to resolve them in a systematic, measurable manner so as to provide effective and optimal results.
Regulation of the Director General Social Forestry and Environmental Partnership Number P.23/PSKL/SET/PSL.3/12/2016 on Guidelines for the Role of Business Actors in the Implementation of Environmental and Forestry Protection and Management	Regulations serve as guidelines and references in implementing the role of the business world in protecting and managing the environment and forestry for governments, companies, communities/communities and related parties.
Regulation of the Director General of Sustainable Production Forest Management Number P. 3/PHPL/SET /Kum.0/2/2017 on Guidelines for Periodic Forest Carbon Inventory (IKHB) in Businesses to Utilize Carbon Absorption and/or Storage in Production Forests	Regulations regulate guidelines for the Periodic Forest Carbon Inventory (IKHB) in Businesses to Utilize Carbon Absorption and/or Storage in Production Forests, the implementation is carried out every 10 years and the results will later become the basis for preparing a Business Work Plan (RKUPHHK).
Regulation of the Director General of Climate Change Control Number P.3/PPI/SET/KUM.1/1/2018 on the Establishment and Development of a Fire Concerned Community	The regulations serve as guidelines and directions for the management unit in the implementation of the formation and development of Fire Concerned Communities (MPA) so that they can be carried out effectively and efficiently, covering the activities of forming, increasing knowledge and skills, independence, and capacity of MPA members
Regulation of the Director General of Climate Change Control Number P.	Regulations serve as guidelines/reference for data collection and reporting on forest and land fire

Regulation Name	Relevance to the Project
8/PPI/PKHL/PPI.4/10/2018 on Guidelines for Reporting Control of Forest and Land Fires	control for management units in a coordinated, integrated, effective, accurate and accountable manner.
Circular Letter of the Director General of Sustainable Production Forest Management Nomor SE.1/PHPL/IPHH/HPL.4/1/2020 on Implementation of GANIS PHPL Determination	The letter regulates determining the Technical Staff for Sustainable Production Forest Management (GANIS-PHPL) based on the GANIS PHPL professional certification obtained through a professional competency test and will later be given a registration number through the Forest Management Technical Personnel Information System (SIGANISHUT)
Circular Letter of the Ministry of Environment and Forestry Number SE.6/MenLHK/Setjen/KUM.1/12/2022 on Approval of Work Plans for Multi-Enterprise Forest Utilization Businesses	This Letter serves as a guideline for implementation in the approval of the Multi-Enterprise Forest Utilization Business Work Plan for Forest Management Business Licenses (PBPH)
Circular Letter of Director General of Sustainable Production Forest Management Bumber SE.7/PHL/PUPH/HPL.1/10/2022 on Protection of Protected Wildlife in the Working Area of Forest Utilization Permits (PBPH)	The letter provides direction to Forest Management Business License holders (PBPH) to further intensify the efforts in protecting the protected wild animals in the PBPH area.
Standard Operational Procedure Number SOP.1/KSDAE/SET.3/KSA.2/12/2022 concerning Wildlife Protection Inside and Outside Forest Areas	The Standard Operational Procedure (SOP) is an effort to protect wild animals as state assets which are one of the potential biological natural resources and ecosystems, protecting them from threats and human disturbance as well as negative interactions between humans and wild animals that occur in forest areas and outside forest areas.

2.5.2 Relevant Laws and Regulations Related to Worker's Rights (VCS, 3.18.2; CCB, G3.11)

This section is still under development

2.5.3 Human Rights (VCS, 3.19)

This section is still under development

2.5.4 Indigenous Peoples and Cultural Heritage (VCS, 3.18, 3.19)

This section is still under development

2.5.5 Statutory and Customary Property Rights (VCS, 3.18, 3.19; CCB, G5.1)

The project area has 2 (two) forest utilization permits, namely for production forest utilization and protected forest utilization. PBPH in production forests is regulated in SK Number SK.782/Menlhk/Setjen/HPL.0/9/2021 dated 16 September 2021 concerning SK Permits for Forest Utilization (PBPH) PT TMK. Some indigenous peoples have been recognized in the project location based on SK.5474/Menlhk-PSKL/PKTHA/PSL.1/10/2020 dated 1 October 2022 concerning Designation of the Mului Indigenous Forest to the Mului Customary Law Community with an area of ±7,722 Ha in Dusun Mului, Swan Slutung Village, Muara Komam District, Paser Regency, East Kalimantan Province.

Participatory mapping with local communities has been carried out before the project started. This is done to determine the boundaries of settlement areas, agricultural areas, customary boundaries, and areas of access to resources related to the project. Communities and stakeholders are also aware of project activities in its management, conservation, and enrichment planting. All statutory provisions related to this project area can be seen in section 2.5.8.

2.5.6 Recognition of Property Rights (VCS, 3.7, 3.18, 3.19; CCB, G5.1)

This section is still under development

2.5.7 Free, Prior and Informed Consent (VCS, 3.18; CCB, G5.2)

This section is still under development

2.5.8 Benefit Sharing Mechanisms (VCS, 3.18, 3.19;)

This section is still under development

2.5.9 Property Rights Protection (VCS, 3.18, 3.19; CCB, G5.3)

This section is still under development

2.5.10 Illegal Activity Identification (VCS, 3.19; CCB, G5.4)

This section is still under development

2.5.11 Ongoing Disputes (VCS, 3.18, 3.19; CCB, G5.5)

This section is still under development

2.5.12 Approvals (CCB, G5.7)

Table 19. Document of Approvals for GLLP

Dokumen	Issued by	Approval Purpose
Decree of the Ministry of Agriculture Number 370/Kpts/Um/7/1970 dated 17 July 1970, PT TMK was granted Natural Forest Concession Rights covering an area of $\pm$ 150,000 Ha as amended by Decree of the Ministry of Agriculture Number 396/Kpts/Um/6/1978 dated 20 June 1978 covering an area of $\pm$ 167,500 Ha in the Province of East Kalimantan for a period of 20 years from 17 July 1970 and ending on 16 July 1990	Ministry of Agriculture	SK IUPHHK
Decree of the Ministry of Forestry No. 478/Kpts-III/1994 dated October 25, 1994, PT TMK received an extension of the Forest Concession Rights for an area of $\pm$ 124,675 Ha for a period of 20 years from July 17, 1990 and ended on July 16, 2010	Ministry of Forestry	Extension of SK IUPHHK
Decree of the Ministry of Environment and Forestry No. SK.34/MenLHK/Setjen/HPL.0/1/2020 concerning Extension of Business Permit for Utilization of Timber Forest Products in Natural Forest on behalf of PT TMK in the area of KPH Telake and KPH Bongan covering an area of $\pm$ 82,805 Ha in Paser Regency, East Kalimantan Province	Ministry of Environment and Forestry Republic of Indonesia	Official license for the rights to utilize timber product in the licensed area
Decree of the Ministry of Environment and Forestry Number SK.782/MenLHK/Setjen/HPL.0/9/2021 dated 16 September 2021 concerning Decree on Forest Utilization Permits (PBPH) for PT TMK covering an area of 82,805 Ha.	Ministry of Environment and Forestry Republic of Indonesia	Forest Utilization Business Licensing Decree
Decree of the Ministry of Environment and Forestry Number SK.7631/MenLHK-PHL/PUPH/HPL.1/9/2022 concerning Approval of Work Plans for Forest Utilization Business Permits for Forest Utilization Period 2022-2031 on behalf of PT TMK, East Kalimantan Province	Ministry of Environment and Forestry Republic of Indonesia	RKUPH Ratification Decree for the 2022-2031 period
Report on the Results of The Boundary Demarcation for Forest Utilization in Natural Forests (PBPH-HA) concession in production forest areas on behalf of PT TMK Number: S.363/BPKHTL.IV/PPKH/PLA.2/5/2023	Ministry of Environment and Forestry Republic of Indonesia	Report and Map of Boundary Demarcation Result
Ministerial Decree for approval revision of the Forest Utilization Business Work Plan (RKUPH) for PT TMK. Number: SK.11658/MenLHK-PHL/PUPH/HPL.1/11/2023	Ministry of Environment and Forestry Republic of Indonesia	Ratification approval decree for revision of the Forest Utilization Business Work Plan (RKUPH)

Dokumen	Issued by	Approval Purpose
		revision for PT TMK
Forest Utilization Business Work Plan (RKUPH) 2022-2031 on behalf of PT TMK initial extension decree (SK.34/Menlhk/Setjen/HPL.0/1/2020 dated 14 January 2020) and the subsequent adjustment decree (SK.782/MENLHK/SETJEN/HPL.0/9/2021 dated 16 September 2021).	Ministry of Environment and Forestry Republic of Indonesia	Ratified revision RKUPH Document for the 2022–2031 period
Legal Basis of Gunung Lumut: Letter Number 503/0022/PBPH/DPMPTSP/II/2023 concerning Recommendations for Applications for Forest Management Business Licenses (PBPH) for PT TMK in Paser Regency, East Kalimantan Province	Provincial Government of East Kalimantan Investment Service and One-Stop Services ((DPMPT)	PBPH HL recommendation on behalf of PT TMK.
Approval of Licensing Commitment to Undertake Forest Utilization in Protected Forest a.n. PT TMK in Paser Regency, East Kalimantan Province Number : 10082311116401003	The Ministry of Environment and Forestry Republic of Indonesia	PBPH HL License Approval on behalf of PT TMK
Official Report In The Making of Geographical Coordinates For The Boundaries of Prospective Work Area Permit In Production Forest PT TMK, Paser Regency, East Kalimantan Province Number: BA.47/BPHL.XI/P3HPL/KG/9/2023	The Ministry of Environment and Forestry Republic of Indonesia	Approval of geographical coordinates for the boundaries of prospective area permit on behalf of PT TMK.

2.5.13 Double Counting and Participation under Other GHG Programs (VCS, 3.23; CCB G5.9)

2.5.13.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program, or any other form of community, social, or biodiversity unit or credit?

☐ Yes ☒ No

2.5.13.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

Is the project active under the other program?

☐ Yes ☒ No

2.5.13.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

2.5.14 Double Claiming, Other Forms of Credit, and Scope 3 Emissions (VCS, 3.24)

2.5.14.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

2.5.14.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

2.5.14.3 Supply Chain (Scope 3) Emissions

Do the project activities affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

2.6 Additional Information Relevant to the Project

2.6.1 Leakage Management (VCS, 3.11, 3.15)

This section is still under development

2.6.2 Further Information

This section is still under development

3 CLIMATE

3.1 Application of Methodology

3.1.1 Title and Reference of Methodology (VCS, 3.1)

Type (methodology, tool, module)	Reference ID (if applicable)	Title	Version
Methodology	VM0007	REDD+ Methodology Framework	1.8
Methodology	VM0010	Improved Forest Management: Conversion From Logged To Protected Forest	1.4
Methodology	VM0048	Reducing Emissions From Deforestation And Forest Degradation	1.0
Methodology	VM0047	Afforestation, Reforestation, And Revegetation	1.1
Module	VMD0001	Estimation of Carbon Stocks in Above- and Belowground Biomass in Live Tree and Non-tree Pools (CP-AB)	1.2
Module	VMD0002	Estimation of Carbon Stocks in the Dead Wood Pool (CP-D)	1.1
Module	VMD0003	Estimation of Carbon Stocks in the Litter Pool (CP-L)	1.1
Module	VMD0004	Estimation of Carbon Stocks in the Soil Organic Carbon Pool (CP-S)	1.1
Module	VMD0005	Estimation of Carbon Stocks in the Long-term Wood Products Pool (CP-W)	1.1
Module	VMD0006	Estimation Of Baseline Carbon Stock Changes And Greenhouse Gas Emissions From Planned Deforestation/Forest Degradation And Planned Wetland Degradation (BI-PI)	1.4

Module	VMD0009	Estimation of emissions from activity shifting for avoiding planned deforestation/forest degradation and avoiding planned wetland degradation (LK-ASP)	1.3
Module	VMD0011	Estimation of Emissions from Market-effects (LK-ME)	1.2
Module	VMD0013	Estimation of Greenhouse Gas Emissions from Biomass and Peat Burning (E BPB)	1.3
Module	VMD0014	Estimation of Emissions from Fossil Fuel Combustion (E-FFC)	1.1
Module	VMD0015	Methods for monitoring of greenhouse gas emissions and removals in REDD project activities (M-REDD)	2.3
Module	VMD0016	Methods for Stratification of the Project Area (X-STR)	1.3
Module	VMD0017	Estimation of uncertainty for REDD+ project activities (X-UNC)	2.2
Module	VMD0019	Methods to Project Future Conditions	1.0
Module	VMD0054	Module for Estimating Leakage from ARR Activities	1.0
Module	VMD0055	Estimation Of Emission Reductions From Avoiding Unplanned Deforestation	1.1
Tool	VT0001	Tool for the Demonstration and Assessment of Additionality in VCS AFOLU Project Activities	3.0
Tool	VT0002	Tool for the Demonstration and Assessment of Additionality in IFM Project Activities Sectoral Scope 14	1.0
Tool		VCS AFOLU Non-Permanence Risk Tool (T-BAR)	4.2

3.1.2 Applicability of Methodology (VCS, 3.1)

VM0007 – REDD+ Methodology Framework

Reference ID/Title	Applicability condition	Justification of conformance
VM0007 (Applicability Conditions)	Land in the project area has qualified as forest (following the definition used by the VCS; in addition, see Section 5.1.2) for at least the 10 years prior to the project start date.	Conditions met. The history of forest cover shows that all land within project area has been covered by tropical forest for at least 10 years.
	Where land within the project area is peatland or tidal wetlands and emissions from the SOC pool are deemed significant, the relevant WRC modules (see Table 3) are applied alongside other relevant modules.	Not applicable. Land within the project area is not peatland or any other wetlands.
	Baseline deforestation in the project area falls within either of the following categories: a) Unplanned deforestation (VCS category AUDef) b) Planned deforestation (VCS category APDef)	Baseline deforestation in the project area falls within the planned deforestation category.
	REDD activity types are not applicable under the following condition: 4) Leakage prevention activities include: a) Flooding agricultural lands to increase production (e.g., rice paddies); and/or b) Intensifying livestock production through use of feed-lots lagoons.	The REDD activity type in the project does not flood agricultural lands to increase production not intensify livestock production.
	Avoiding planned deforestation activities are applicable under the following condition: 7) Where conversion of forest lands to a deforested condition is legally permitted	Conditions met. The land use scenario for the baseline (coal mining) is legally permitted.

VM0010 – Improved Forest Management : Logged to Protected Forest

Reference ID/Title	Applicability condition	Justification of conformance
VM0010 (Applicability Conditions)	Forest management in the baseline scenario is planned selective logging.	Justification that the project conforms with this applicability condition
...	Planned selective logging must be estimated using forest inventory methods that determine allowable offtake as volume of timber (m3 Ha-1).	The volume of wood stock that can be harvested (D>40 cm) per unit area has been estimated using field measurement data based on the results of a forest inventory using IHMB (Ministry of Environment Regulation No. 8 of 2021), which is carried out in an integrated manner with carbon inventory
	Boundaries of the forest land are clearly defined and documented.	The forest land has been clearly defined, delineated, and documented
	All applicability conditions of VCS and CDM tools used in conjunction with this methodology are met	Conditions met. All applicability conditions of VCS and CDM tools used in conjunction with this methodology are met
Legal Right to Harvest	The legal right to harvest must pre-exist the implementation of the project.	Under the baseline scenario, the project area is being managed as a production forest, which has been designated for logging
	The legal right to harvest must be issued by a relevant government body, define a legal allocation of rights to a forest timber resource, and includes a plan for forest management that contains a definition of the spatial extent of the forest, the volume of the timber resource to be extracted and a description of harvesting practices.	The GLLP Unit I project area is an area legally designated as a Business Permit for the Utilization of Timber Forest Products in Natural Forests ("IUPHHK-HA") based on Decree Number: SK.34/Menlhk/Setjen/HPL.0/1/2020 dated January 14, 2020 concerning Extension of IUPHHK-HA on behalf of PT TMK in the area of KPH Telake and KPH Bongan covering an area of ± 82,805 Ha in Paser Regency, East Kalimantan Province. Evidence and legality of the right to harvest timber are issued by the relevant government agency, which has designated and approved the project area for forest management, and established the allocation of legal rights to forest timber resources and harvesting practices.

Reference ID/Title	Applicability condition	Justification of conformance
	Rights to forest management must be demonstrated by documentary proof of legal permissibility for timber harvest, intent to harvest and a description of the timber resource. This proof must be issued by the relevant (governmental) regulatory body that has designated, sanctioned or approved the project area (or areas) for forest management.	
Intent to Harvest	- Documented evidence demonstrating that: - The project site is representative of other forestlands harvested in the country within the past two years; and, - The project site is within commercially viable distance to existing transport networks and a port for timber export or a mill for timber processing; or, - A valid and verifiable government-approved timber management plan for harvesting the project area.	Before carrying out the project intervention plan, the project proponent planned to carry out timber harvesting activities through work area arrangement documents (“PAK”) that is listed inside the Environmental Impact Analysis document (“AMDAL”).
VM0010 (not applicable conditions)	The project area includes planted forests.	The project area does not include planted forests.
	Under the project scenario, forest use in the project area includes commercial timber harvest or activities that cause forest degradation.	Under the project scenario, the forest area will be fully protected, and no commercial timber and wood harvesting is carried out.
	The baseline scenario includes conversion to managed plantations.	Based on the permit and regulation, the project area is under Natural Forest Concession, hence, will not be, in any case, converted into managed plantations.
	The project occurs in wetlands or peatlands.	This condition is met, the type of soil in the entire project area is mineral soil. There is no peat soil, water bodies and swamps all of

Reference ID/Title	Applicability condition	Justification of conformance
		which are not included in the calculation of the baseline scenario or project scenario.
	The project occurs in a location where VM0045 is currently applicable.	The project area does not occur where VM0045 is currently applicable.
VMD0004	Project activity is applied on non-organic soils	This condition is met, the type of soil in the entire project area is mineral soil. There is no peat soil, water bodies and swamps all of which are not included in the calculation of the baseline scenario or project scenario

VM0048 – VMD0055 : Reducing Emissions From Deforestation And Forest Degradation

Reference ID/Title	Applicability condition	Justification of conformance
VM0048-AuDef (Applicability Conditions)	The land use transition in the baseline scenario is forest land to non-forest land, meeting the definition of unplanned deforestation;	Conditions met. Based on historical data on deforestation areas in the GLLP Unit II area and region reference, the drivers of deforestation are clearing land for illegal tree harvesting and crop production (farmers)/non-large-scale agriculture.
...	The project involves activities aimed at avoiding unplanned deforestation	Conditions met. The project involves activities aimed at avoiding unplanned deforestation
	Agents of deforestation in the baseline scenario clear the land for tree harvesting, settlements, roads, unsanctioned expansion of roads or other infrastructure, agricultural crop production, ranching, or aquaculture.	Conditions met. Based on historical data on deforestation areas in the GLLP Unit II area and region reference, the drivers of deforestation are clearing land for illegal tree harvesting and crop production (farmers)/non-large-scale agricultur
	The entire UDef PA must be forest at the project start date and must only include land qualifying as forest for a minimum of 10 years prior to the project start date. The boundaries of the UDef PA remain fixed for the duration of the project crediting period, regardless of any land cover	Conditions met. The project area in Unit II is land qualifying as forest for more than 10 years prior to the project start date. The boundaries of the UDef PA will remain fixed for the duration of the project crediting period.

Reference ID/Title	Applicability condition	Justification of conformance
	change after the start of the project.	
VM0048-AuDef (Not Applicable Conditions)	The project harvests trees for wood products (e.g., timber, fuelwood, charcoal, fiber), unless the trees have been demonstrably planted during the project lifetime as part of project activities for leakage mitigation or community development;	Under the project scenario, the forest area will be fully protected, and no commercial timber and wood harvesting is carried out.
	More than two percent of the vegetation cover in the project area is peatland or tidal wetland vegetation;	The type of soil in the entire project area is mineral soil. There is no peat soil, water bodies and swamps all of which are not included in the calculation of the baseline scenario or project scenario
	Wetlands are being drained in the baseline scenario or wetland drainage occurs in the project scenario (including the use of trenches or ditches for the purpose of wetland drainage);	The type of soil in the entire project area is mineral soil. There is no peat soil, water bodies and swamps all of which are not included in the calculation of the baseline scenario or project scenario
	More than ten percent of the vegetation cover in a 10-kilometer-wide buffer around the project area is peatland or tidal wetland vegetation; or	The type of soil in the entire project area and surrounding the project area covering a 10 kilometer wide buffer around the project area is mineral soil.
	Implemented leakage mitigation measures include: a) Flooding agricultural lands to increase production (e.g., for rice paddies), or b) Intensifying livestock production through the use of feed-lots and/or manure lagoons.	The project does not cover agricultural development on flooded lands or intensification of livestock production.

VM0047- Afforestation, Reforestation, And Revegetation

Reference ID/Title	Applicability condition	Justification of conformance
Applicability Conditions: General	Project activities increase vegetative cover.	The project activities will aim to increase vegetative cover in the project area.

Reference ID/Title	Applicability condition	Justification of conformance
	Where area-based and census-based approaches are used together, they are applied in non-overlapping areas defined at the project start	Areas defined in the project start will not have any overlapping issues.
	For lands enrolled in a project, the quantification approach is selected at the project start date and used for the entire project crediting period.	Quantification approach is selected at the project start date and used for the entire project crediting period.
	The project start date is documented as the earliest of the following: a) The date on which site preparation activities began; b) The land use change date	The project start date is documented as the earliest on the date which site preparation activities began.
	Where projects take place on organic soils or wetlands, ARR activities are developed using a multiple project activity design, applying this methodology to account for aboveground biomass and using a Wetland Restoration and Conservation methodology	The project will not take place on organic soils or wetlands.
Applicability Conditions: Area-based	Project activities involve direct planting activities (e.g., manual planting, broadcast seeding), indirect activities associated with assisted natural regeneration (e.g., liana cutting, weed management, or barriers that prevent animal grazing), or a combination of direct and indirect activities.	
	Project proponents establish a $t = 0$ carbon stock estimate for all significant carbon pools. The method for establishing $t = 0$ estimates depends on the activity that initiates the project start date.	

Reference ID/Title	Applicability condition	Justification of conformance
	Where the project start date is initiated by site preparation (e.g., clearing invasive species), the following conditions must be met: i) Before site preparation, project proponents establish $t = 0$ carbon stock estimates for all significant carbon pools ³ . ii) To ensure accuracy, $t = 0$ estimates are established no more than two years before the start date (e.g., if the start date is 2025, plots must have been established no earlier than 2023). iii) Where plot-based measurements are not established before site preparation, a remote sensing-based estimate found in Section 8.2.1.2 may be used, but only for pre-existing woody biomass.	
	Where the project start date is defined by a land use change date or where the project start did not include site preparation that caused a significant decrease in carbon stocks in monitored carbon pools (e.g., pit planting without clearing existing vegetation) the following conditions must be met: i) The project proponent establishes $t = 0$ estimates within two years after the project start date (e.g., if the start date is 2025, plots must be established by 2027). Plot-based sampling occurs for all significant carbon pools. iii) Evidence is provided to demonstrate that site preparation did not involve clearing, burning, or mechanical disturbance of existing vegetation that would significantly reduce monitored carbon pools. Evidence may	

Reference ID/Title	Applicability condition	Justification of conformance
	include, but is not limited to, georeferenced photos, field survey data, satellite imagery, and signed attestations from landowners.	
	Leakage must be monitored and quantified using VMD0054. It must not be assumed to be de minimis.	
Applicability Conditions Census-based approach	The project activity only includes direct planting.	The project activity will be about direct planting on the ARR project activity area.
	The pre-project land use is maintained throughout the project lifetime (e.g., where projects occur on agricultural land, agricultural production continues).	Pre-project land use is maintained throughout the project lifetime.
	Planting density does not exceed 50 planting units per hectare. The planting density limit applies proportionally to the size of each instance: a) Where an instance is less than one hectare or includes part of a hectare (e.g., 0.50 Ha or 1.50 Ha, respectively), the planting density is scaled proportionally to the size of the instance. For example, an instance of 0.50 Ha may include no more than 25 planting units. b) In instances larger than one hectare, planting is dispersed to maintain compliance with the 50 planting units per hectare limit across the entire instance (e.g., in a 10 hectare instance, up to 500 planting units may be planted, but must not be concentrated in a single hectare or portion of the instance).	The planting density will not exceed 50 planting units per hectare.
	The establishment of a complete census of all planting units marks the start of the	

Reference ID/Title	Applicability condition	Justification of conformance
	project and is $t = 0$. The census is conducted to establish N, the total number of planting units. Only planting units planted by the project proponent are included in the census. Existing vegetation is not included in the census and does not count toward the 50 planting units per hectare density threshold.	
	Where planting units have died, plantings may be replanted provided N does not exceed 50 live planting units per hectare established by the project activity.	
	Individual planting units of woody biomass are clearly defined (e.g., tree, shrub, bamboo clump) and identifiable in the field. One of the following identification approaches must be used: a) GPS points: Project proponents must ensure that the spacing between individual planting units is greater than or equal to the positional accuracy of the Global Positioning System (GPS) units used for geolocating each planting unit. For example, if the positional accuracy of the GPS is five meters, the minimum spacing between planting units must be greater than or equal to five meters. b) Physical markers: Each planting unit must be marked with a durable, in-field physical identifier bearing a unique ID. These markers must be clearly visible and easily located during verification activities.	
	Where a planting unit cannot be located during a monitoring event, it is conservatively	

Reference ID/Title	Applicability condition	Justification of conformance
	assumed that the planting unit has died.	
	The project activity occurs: a) within an area with less than 10% pre-existing woody biomass cover; and/or b) in an area subject to continuous cropping, or in “settlements” or “other lands” land use categories.	
	Soil disturbance from project activities ⁶ (e.g., site preparation) meets the following conditions: a) Soil disturbance is permitted only at the time of planting. Where replanting or additional waves of planting occur, soil disturbance is permitted under the same conditions. b) Planting techniques that cause localized soil disturbance, such as pit planting, may exceed a depth of 25 cm. c) Where planting involves soil inversion (e.g., plowing), the depth must not exceed 25 cm (such as that caused by a moldboard plow) and may only occur once during the project crediting period.	
Area-based exclusion	1) The project occurs on lands that have met the definition of managed forest at any point in the 10-year period immediately preceding the project start date. 2) Clearing of pre-existing woody biomass involves timber harvesting or results in degradation of native ecosystems. 3) The project is planting fewer than 50 planting units per hectare and could use the census-based approach.	
Census-based exclusion	1) Woody biomass, which serves a similar purpose as the planting units in the project, has	No biomass have been purposely removed within the past 10 years for planting unit purposes in the project. The project activity

Reference ID/Title	Applicability condition	Justification of conformance
	been removed within the last 10 years (confirmed via pre-project photos and/or attestation). 2) Soil disturbance from the project activity involves soil inversion to a depth exceeding 25 cm (e.g., that would result from a moldboard plow).	will not involve soil inversion to a depth exceeding 25 cm.

3.1.3 Project Boundary (VCS, 3.12)

VM0007 – REDD Methodology Framework for Avoid Planned Deforestation

Source		Gas	Included?	Justification/explanation
Baseline	Burning of woody biomass	CO2	Included	Carbon stock decreases due to burning are accounted as a carbon stock change.
		CH4	Included	Non-CO2 gases emitted from woody biomass burning – it is conservative to exclude.
		N2O	Included	
	Combustion of fossil fuels	CO2	Included	It is conservative to exclude
		CH4	Excluded	Potential emissions are negligible
		N2O	Excluded	Potential emissions are negligible
	Use of fertilizers	CO2	Excluded	Potential emissions are negligible
		CH4	Excluded	Potential emissions are negligible
		N2O	Included	It is conservative to exclude
Project	Burning of woody biomass	CO2	Included	Carbon stock decreases due to burning are accounted as a carbon stock change
		CH4	Included	Non-CO2 gases emitted from woody biomass burning
		N2O	Included	
	Combustion of	CO2	Included	Can be neglected if excluded from baseline accounting
		CH4	Excluded	Potential emissions are negligible

Source		Gas	Included?	Justification/explanation
	fossil fuels	N2O	Excluded	Potential emissions are negligible
	Use of fertilizers	CO2	Excluded	Potential emissions are negligible
		CH4	Excluded	Potential emissions are negligible
		N2O	Included	Can be excluded if excluded from baseline accounting except in the situation where fertilizer use is enhanced as a leakage avoidance mechanism

VM0010 – Improved Forest Management : Logged to Protected Forest

Source		Gas	Included?	Justification/explanation
Baseline & Project	Combustion of fossil fuels (in vehicles, machinery and equipment)	CO2	Included, if significant	Only included if tested as significant. Otherwise excluded, which is deemed conservative, as emissions will be greater in the baseline scenario than in the project scenario
		CH4	Included, if significant	Only included if tested as significant. Otherwise excluded, which is deemed conservative, as emissions will be greater in the baseline scenario than in the project scenario
		N2O	Included, if significant	Only included if tested as significant. Otherwise excluded, which is deemed conservative, as emissions will be greater in the baseline scenario than in the project scenario
	Removal of herbaceous vegetation	CO2	Excluded	Based on CDM EB decision reflected in paragraph 11 of the report of the 23rd session of the board: https://cdm.unfccc.int/Panels/ar/023/ar_023_rep.pdf
	Burning of Biomass	CH4	Included	Included as CO2 equivalent emissions
		N2O	Excluded	Potential emissions are negligible
	Nitrogen based fertilizer	N2O	Excluded	Potential emissions are negligible. Following the VCS update to the Tool for AFOLU Methodological Issues and Guidance for AFOLU Projects, emissions via the use of

Source	Gas	Included?	Justification/explanation
			fertilizer are considered insignificant and are not considered here.

VM0048 – Reducing Emissions From Deforestation And Forest Degradation for Avoid Unplanned Deforestation

Source		Gas	Included?	Justification/explanation
Baseline	Burning of woody biomass	CO2	Included	Major emissions source
		CH4	Optional	Non-CO2 gases emitted from woody biomass burning – it is conservative to exclude.
		N2O	Optional	
	Combustion of fossil fuels	CO2	Optional	May be excluded where determined negligible
		CH4	Excluded	Potential emissions are negligible.
		N2O	Excluded	Potential emissions are negligible.
	Use of fertilizers	CO2	Excluded	Potential emissions are negligible.
		CH4	Excluded	Potential emissions are negligible.
		N2O	Optional	May be excluded where determined negligible
Project	Burning of woody biomass	CO2	Included	Major emissions source
		CH4	Included	Major emissions source
		N2O	Included	Major emissions source
	Combustion of fossil fuels	CO2	Optional	Emissions associated with the combustion of fossil fuels due to leakage prevention activities are always considered insignificant. Emissions associated with other activities (e.g., monitoring, patrolling) must be demonstrated as negligible to be omitted.
		CH4	Excluded	Potential emissions are negligible.
		N2O	Excluded	Potential emissions are negligible.
		CO2	Excluded	Potential emissions are negligible.

Source	Gas	Included?	Justification/explanation
Use of fertilizers	CH ₄	Excluded	Potential emissions are negligible.
	N ₂ O	Optional	Must be included where fertilizer use increases due to the project (e.g., as a leakage avoidance mechanism). Otherwise, it may be excluded if it is also excluded from the baseline.

VM0047 – Afforestation, Reforestation, And Revegetation

Source		Gas	Included?	Justification/explanation	
Baseline	Burning of biomass (natural or anthropogenic causes)	CO2	Excluded	Conservative to exclude	
		CH4	Excluded	Conservative to exclude	
		N2O	Excluded	Conservative to exclude	
	Emissions from nitrogen fertilizer	CO2	Excluded	Conservative to exclude	
		CH4	Excluded	Conservative to exclude	
		N2O	Excluded	Conservative to exclude	
	Burning of fossil fuels	CO2	Excluded	Conservative to exclude	
		CH4	Excluded	Conservative to exclude	
		N2O	Excluded	Conservative to exclude	
	Project	Burning of biomass (natural or anthropogenic causes)	CO2	Excluded	Carbon stock decreases due to burning are accounted as a carbon stock change.
			CH4	Included	May be a significant source.
			N2O	Included	May be a significant source.
Emissions from		CO2	Excluded	Conservative to exclude	
		CH4	Excluded	Conservative to exclude	

Source		Gas	Included?	Justification/explanation
	nitrogen fertilizer	N2O	Included	May be a significant source.
	Burning of fossil fuels	CO2	Excluded	De minimis
		CH4	Excluded	De minimis
		N2O	Excluded	De minimis

3.1.4 Baseline Scenario (VCS, 3.13)

This section is still under development

3.1.5 Additionality (VCS, 3.14)

This section is still under development

3.1.5.1 Regulatory Surplus (VCS, 3.14)

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes ☒ No

3.1.5.2 Additionality Methods (VCS, 3.14)

This section is still under development

3.1.6 Methodology Deviations (VCS, 3.20)

This section is still under development

3.2 Quantification of Estimated GHG Emission Reductions and Removals

3.2.1 Baseline Emissions (VCS, 3.15)

This section is still under development

3.2.2 Project Emissions (VCS, 3.15)

This section is still under development

3.2.3 Leakage Emissions (VCS 2.5, 3.2, 3.6, 3.15, 4.3)

This section is still under development

3.2.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals (VCS, 3.15, 4.1)

This section is still under development

3.3 Monitoring

3.3.1 Data and Parameters Available at Validation (VCS, 3.16)

This section is still under development

3.3.2 Data and Parameters Monitored (VCS, 3.16)

This section is still under development

3.3.3 Monitoring Plan (VCS, 3.16, 3.20)

This section is still under development

3.3.4 Dissemination of Monitoring Plan and Results (VCS, 3.18; CCB, CL4.2)

This section is still under development

3.4 Optional Criterion: Climate Change Adaptation Benefits

3.4.1 Regional Climate Change Scenarios (CCB, GL1.1)

This section is still under development

3.4.2 Climate Change Impacts (CCB, GL1.2)

This section is still under development

3.4.3 Measures Needed and Designed for Adaptation (CCB, GL1.3)

This section is still under development

4 COMMUNITY

4.1 Without-Project Community Scenario

4.1.1 Descriptions of Communities at Project Start (CCB, CM1.1)

This section is still under development

4.1.2 Interactions between Communities and Community Groups (VCS, 3.19; CCB, CM1.1)

This section is still under development

4.1.3 High Conservation Values (CCB, CM1.2)

This section is still under development

4.1.4 Without-Project Scenario: Community (CCB, CM1.3)

This section is still under development

4.2 Net Positive Community Impacts

4.2.1 Expected Community Impacts (CCB, CM2.1)

This section is still under development

4.2.2 Negative Community Impact Mitigation (VCS, 3.19; CCB, CM2.2)

This section is still under development

4.2.3 Net Positive Community Well-Being (VCS, 3.19; CCB, CM2.3, GL1.4)

This section is still under development

4.2.4 High Conservation Values Protected (CCB, CM2.4)

This section is still under development

4.3 Other Stakeholder Impacts

4.3.1 Impacts on Other Stakeholders (VCS, 3.18, 3.19; CCB, CM3.1)

This section is still under development

4.3.2 Mitigation of Negative Impacts on Other Stakeholders (VCS, 3.18, 3.19; CCB, CM3.2)

This section is still under development

4.3.3 Net Impacts on Other Stakeholders (VCS, 3.18, 3.19; CCB, CM3.3)

This section is still under development

4.4 Community Impact Monitoring

4.4.1 Community Monitoring Plan (CCB, CM4.1, CM4.2, GL1.4, GL2.2, GL2.3, GL2.5)

This section is still under development

4.4.2 Monitoring Plan Dissemination (CCB, CM4.3)

This section is still under development

4.5 Optional Criterion: Exceptional Community Benefits

This section is still under development

4.5.1 Exceptional Community Criteria (CCB, GL2.1)

This section is still under development

4.5.2 Short-term and Long-term Community Benefits (CCB, GL2.2)

This section is still under development

4.5.3 Community Participation Risks (CCB, GL2.3)

This section is still under development

4.5.4 Marginalized and/or Vulnerable Community Groups (CCB, GL2.4)

This section is still under development

4.5.5 Net Impacts on Women (CCB, GL2.5)

This section is still under development

4.5.6 Benefit Sharing Mechanisms (CCB, GL2.6)

This section is still under development

4.5.7 Benefits, Costs, and Risks Communication (VCS, 3.18; CCB, GL2.7)

This section is still under development

4.5.8 Governance and Implementation Structures (CCB, GL2.8)

This section is still under development

4.5.9 Smallholders/Community Members Capacity Development (CCB, GL2.9)

This section is still under development

5 BIODIVERSITY

5.1 Without-Project Biodiversity Scenario

5.1.1 Existing Conditions (VCS, 3.19; CCB, B1.1)

This section is still under development

5.1.2 High Conservation Values (CCB, B1.2)

This section is still under development

5.1.3 Without-project Scenario: Biodiversity (CCB, B1.3)

This section is still under development

5.2 Net Positive Biodiversity Impacts

5.2.1 Expected Biodiversity Changes (VCS, 3.19; CCB, B2.1)

This section is still under development

5.2.2 Mitigation Measures (VCS, 3.19; CCB, B2.3)

This section is still under development

5.2.3 Net Positive Biodiversity Impacts (CCB, B2.2, GL1.4)

This section is still under development

5.2.4 High Conservation Values Protected (CCB, B2.4)

This section is still under development

5.2.5 Species Used (VCS, 3.19; CCB, B2.5, B2.6)

This section is still under development

5.2.6 Invasive Species (VCS, 3.19; CCB, B2.5)

This section is still under development

5.2.7 GMO Exclusion (CCB, B2.7)

This section is still under development

5.2.8 Inputs Justification (VCS, 3.19; CCB, B2.8)

This section is still under development

5.2.9 Waste Products (VCS, 3.19; CCB, B2.9)

This section is still under development

5.3 Offsite Biodiversity Impacts

5.3.1 Negative Offsite Biodiversity Impacts (CCB, B3.1) and Mitigation Measures (CCB, B3.2)

This section is still under development

5.3.2 Net Offsite Biodiversity Benefits (VCS, 3.19; CCB, B3.3)

This section is still under development

5.4 Biodiversity Impact Monitoring

5.4.1 Biodiversity Monitoring Plan (CCB, B4.1, B4.2, GL1.4, GL3.4)

This section is still under development

5.4.2 Biodiversity Monitoring Plan Dissemination (CCB, B4.3)

This section is still under development

5.5 Optional Criterion: Exceptional Biodiversity Benefits

This section is still under development

5.5.1 High Biodiversity Conservation Priority Status (CCB, GL3.1)

This section is still under development

5.5.2 Trigger Species Population Trends (CCB, GL3.2, GL3.3)

This section is still under development

APPENDIX 1: STAKEHOLDER DESCRIPTION TABLE

This section is still under development

APPENDIX 2: PROJECT ACTIVITIES AND THEORY OF CHANGE TABLE

This section is still under development

APPENDIX 3: PROJECT RISKS TABLE

This section is still under development

APPENDIX 4: COMMERCIALY SENSITIVE INFORMATION

This section is still under development